Comparison Between RS67 Vs RS57-70

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May 5, 2026

Slide 02: Input Files (Roadset & Target Order)

- merged_RS57_LH2_1_1138.root
- merged_RS57_LD2_3_1138.root
- merged_RS57_Empty_2_1138.root
- merged_RS59_LH2_1_465.root
- merged_RS59_LD2_3_466.root
- merged_RS59_Empty_2_466.root
- merged_RS62_LH2_1_1234.root
- merged_RS62_LD2_3_1237.root
- merged_RS62_Empty_2_1234.root
- merged_RS67_3089LH2.root
- merged_RS67_3089LD2.root
- merged_RS67_3089flask.root
- merged_RS70_LH2_1_264.root
- merged_RS70_LD2_3_266.root
- merged_RS70_Empty_2_267.root

Remark: Currently we don't process RS5a, RS5b files.

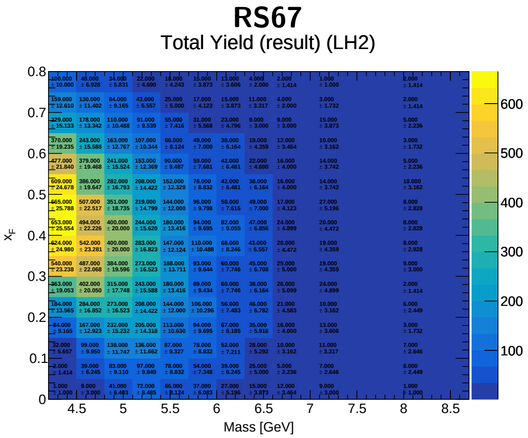
Slide 03: Physics Constants & Normalizations

Parameter	Value
PoT LH_2	2.7639×10^{17}
PoT LD_2	1.3197×10^{17}
PoT Flask	5.5905×10^{16}
LH_2 Target Density	3.5966 mol/cm^2
LD_2 Target Density	8.0431 mol/cm^2
LH_2 / LD_2 Target Length	50.8 cm
Avogadro Constant	6.022×10^{23}
xF Bin Width	0.05
Nuclear Int. Length (LH_2/LD_2)	52.0 / 54.7 g/cm^2

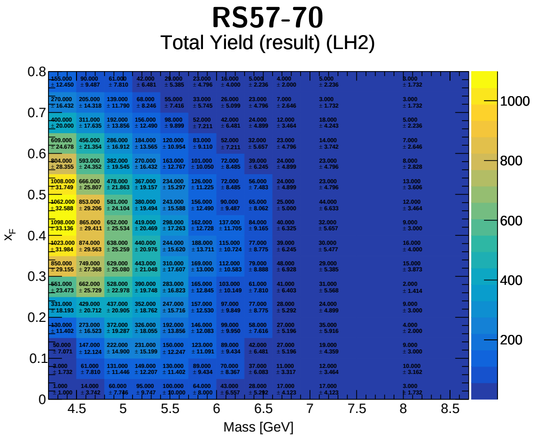
Slide 04: Event Selection Summary

Dimuon Level	$\|dx\| < 0.25$, mass $\in [4.2, 8.8]$, $xF \in [-0.1, 0.95]$, $xT \in [0.05, 0.58]$, $\chi^2 < 18$.
Track Level	$\chi^2_{target} < 15$, $pz_{st1} \in [9, 75]$, Hits > 13, Station 1/3 y-ratio < 1.
Occupancy	$D1, D2, D3 < 400$ hits each; Total $D1 + D2 + D3 < 1000$.
Physics	$Z_{vertex} \in [-320, -5]$, Track Separation < 270, $\|\cos\theta\| < 0.5$.

Y_total_LH2

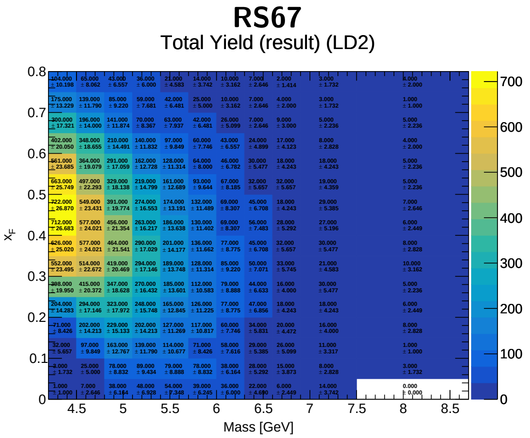


RS67
Total Yield (result) (LH2)
Mass [GeV]
x_F

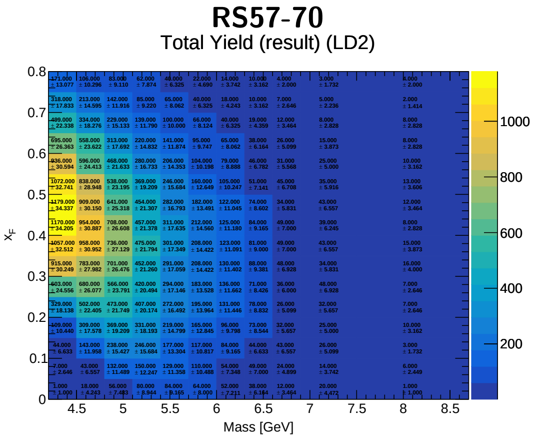


RS57-70
Total Yield (result) (LH2)
Mass [GeV]
x_F

Y_total_LD2

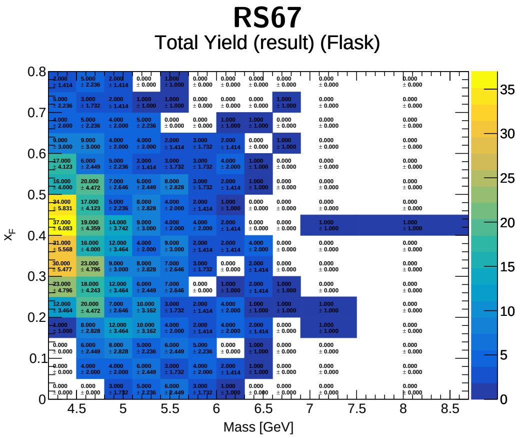


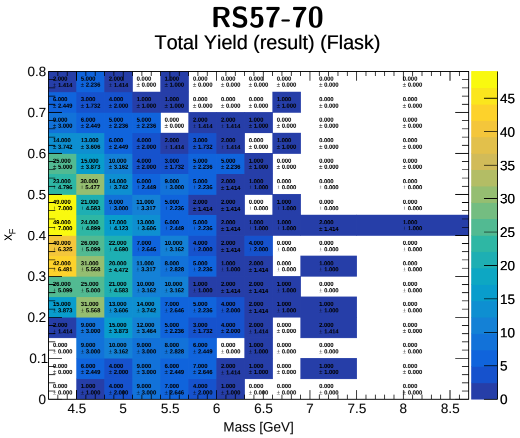
RS67
Total Yield (result) (LD2)
x_F
Mass [GeV]



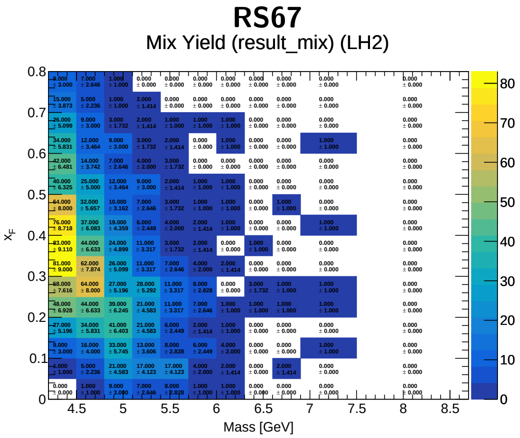
RS57-70
Total Yield (result) (LD2)
x_F
Mass [GeV]

Y_total_Flask

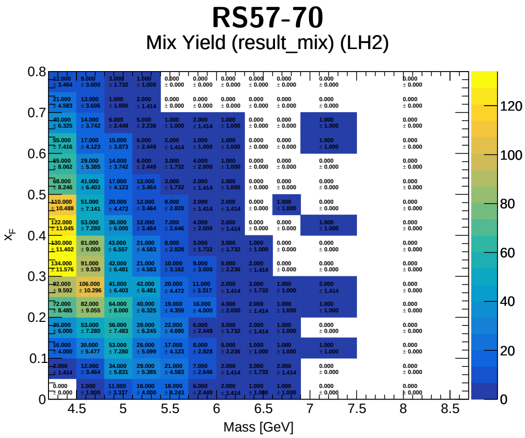




Y_mix_LH2

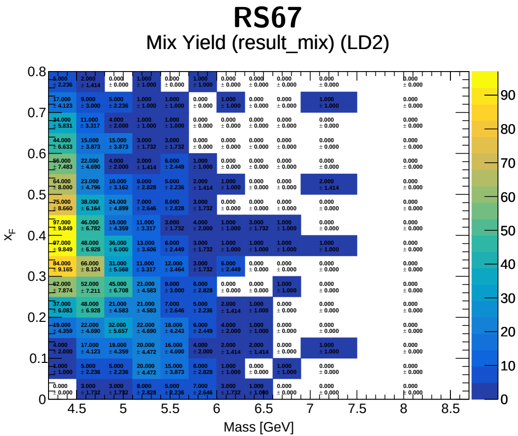


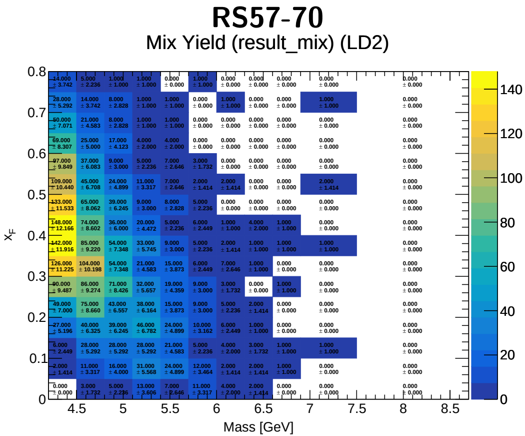
RS67
Mix Yield (result_mix) (LH2)
x_F
Mass [GeV]



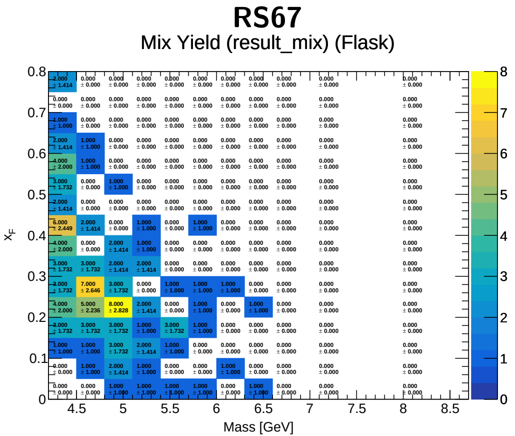
RS57-70
Mix Yield (result_mix) (LH2)
x_F
Mass [GeV]

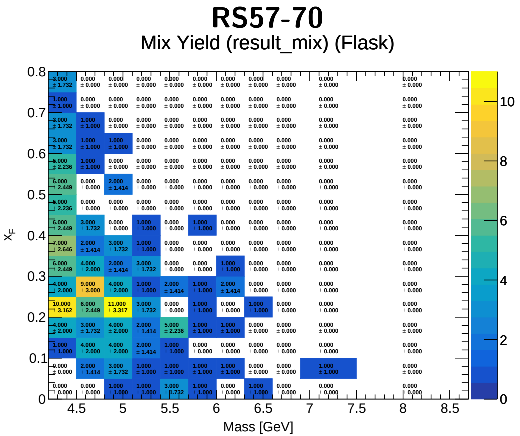
Y_mix_LD2


RS67
Mix Yield (result_mix) (LD2)
x_F
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
4.5
5
5.5
6
6.5
7
7.5
8
8.5
Mass [GeV]
90
80
70
60
50
40
30
20
10
0

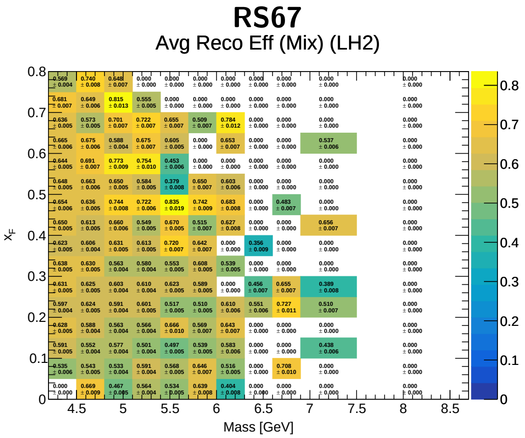

RS57-70
Mix Yield (result_mix) (LD2)
x_F
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
4.5
5
5.5
6
6.5
7
7.5
8
8.5
Mass [GeV]
140
120
100
80
60
40
20
0

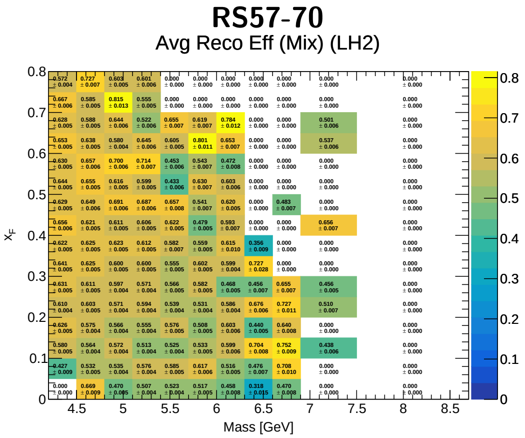
Y_mix_Flask


RS67
Mix Yield (result_mix) (Flask)
x_F
Mass [GeV]

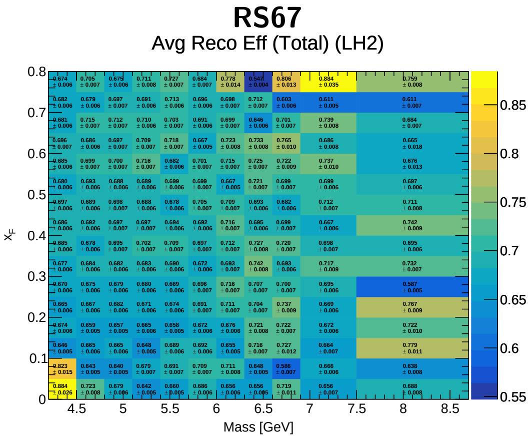

RS57-70
Mix Yield (result_mix) (Flask)
x_F
Mass [GeV]

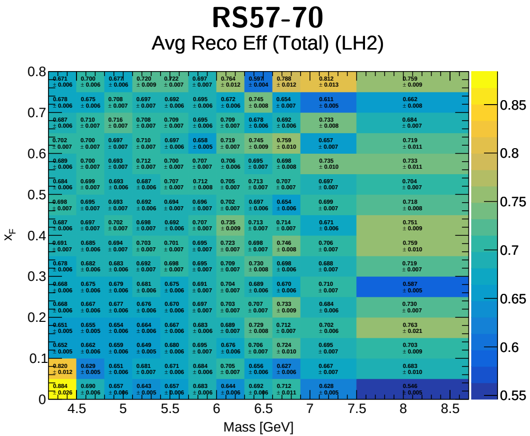
E_mix_reco_LH2



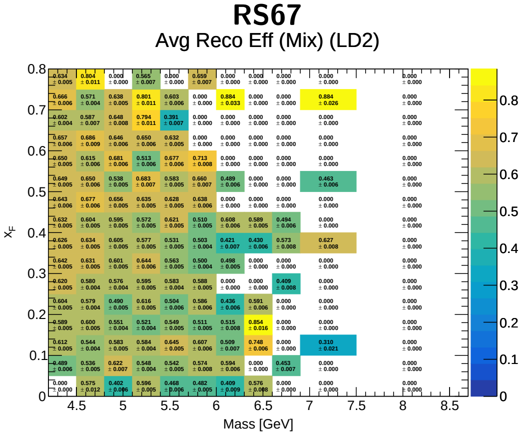


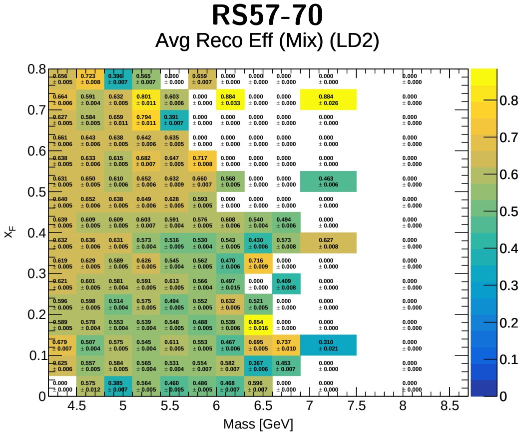
E_total_reco_LH2



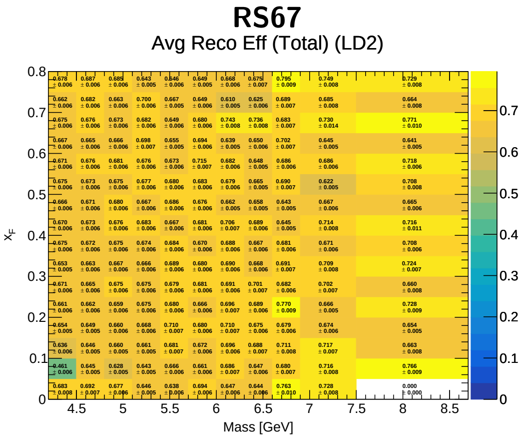


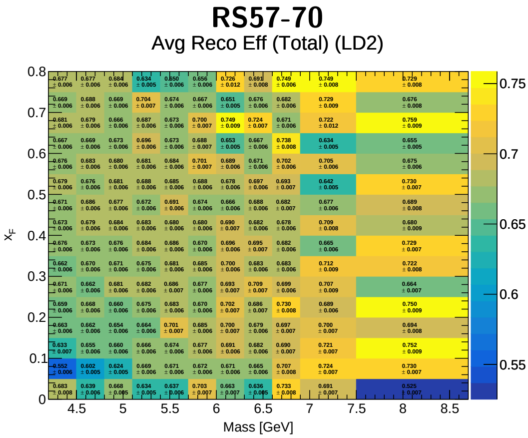
E_mix_reco_LD2



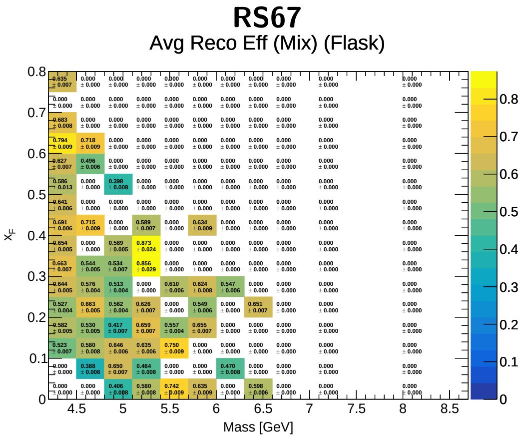


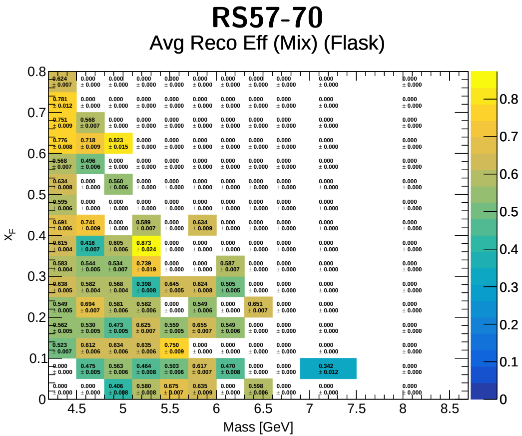
E_total_reco_LD2



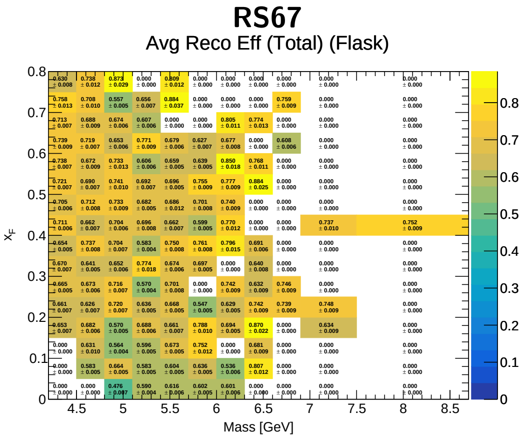


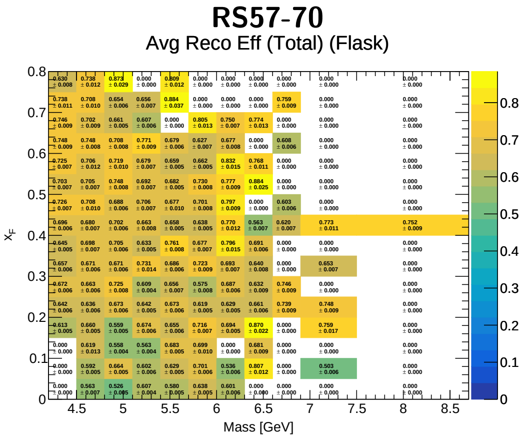
E_mix_reco_Flask

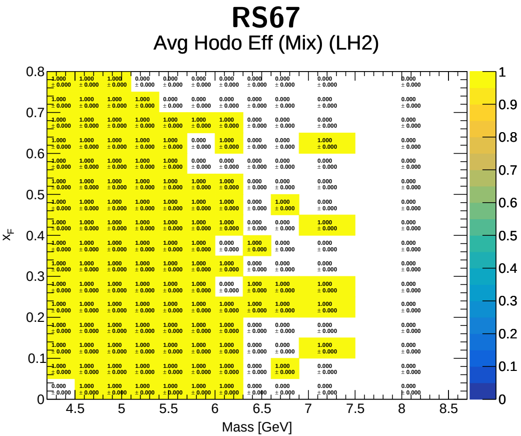


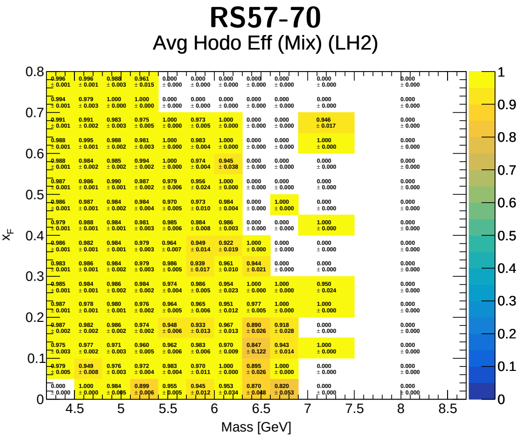


E_total_reco_Flask

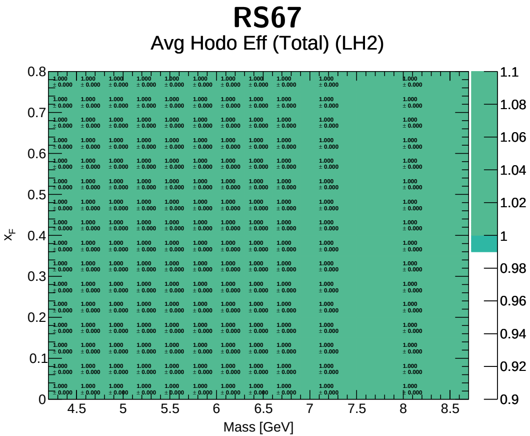


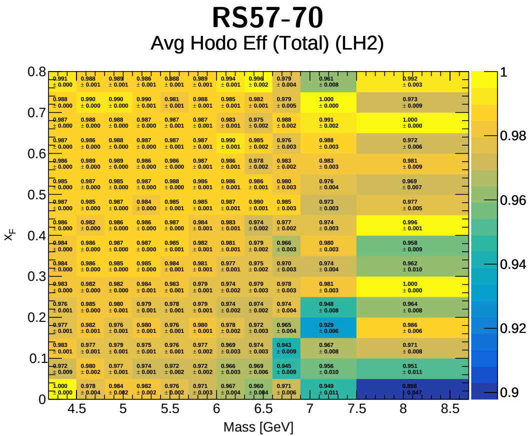



RS67
Avg Hodo Eff (Mix) (LH2)
x_F
Mass [GeV]

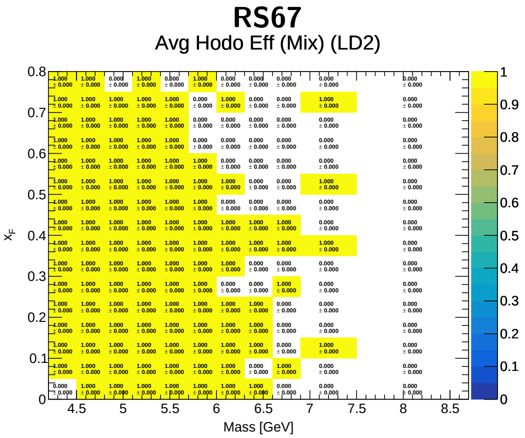

RS57-70
Avg Hodo Eff (Mix) (LH2)
x_F
Mass [GeV]

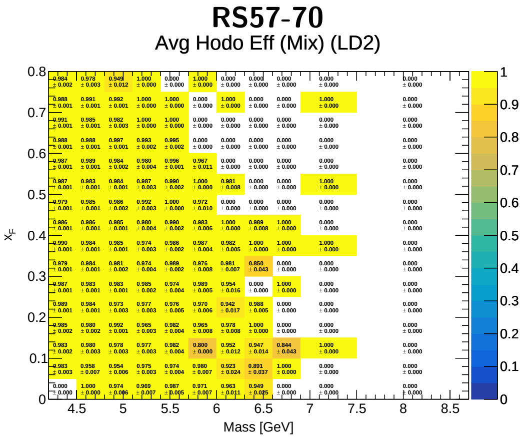
E_total_hodo_LH2



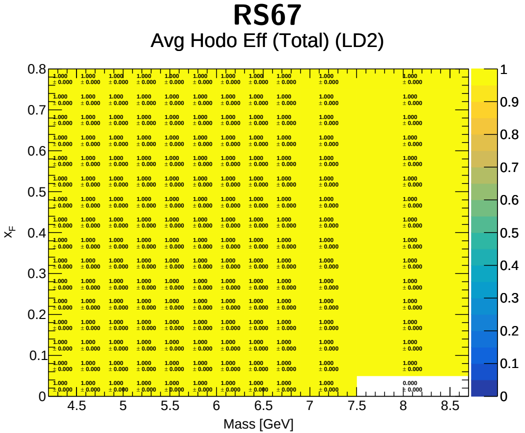


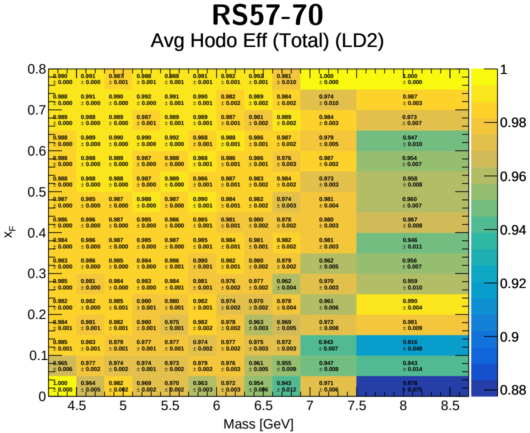
E_mix_hodo_LD2



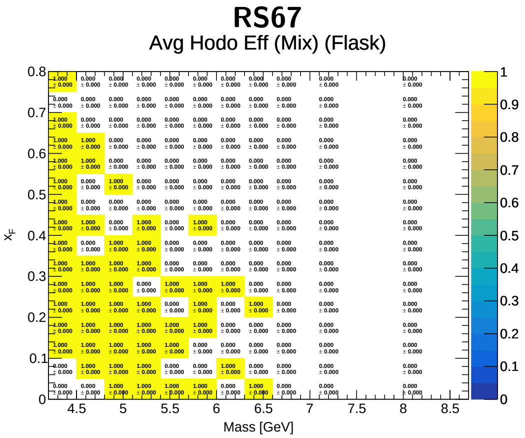


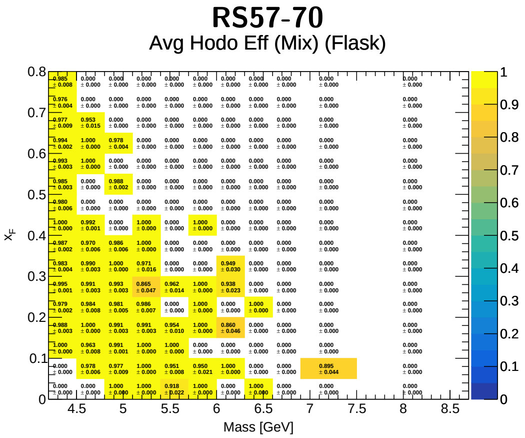
E_total_hodo_LD2





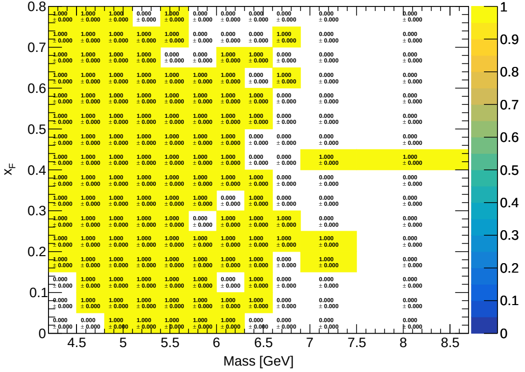
E_mix_hodo_Flask



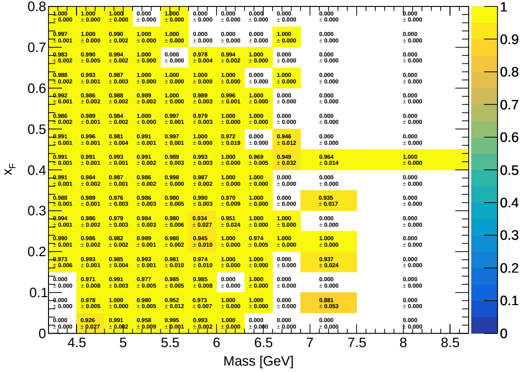


E_total_hodo_Flask

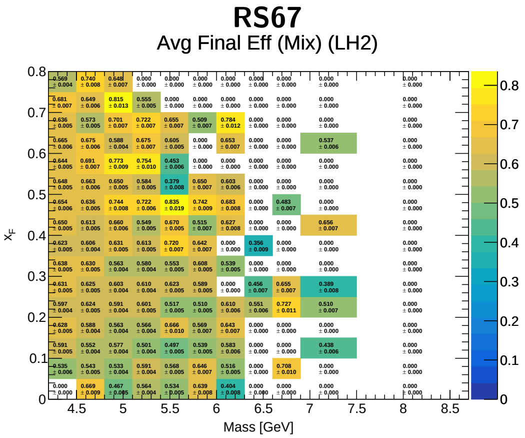
RS67
Avg Hodo Eff (Total) (Flask)

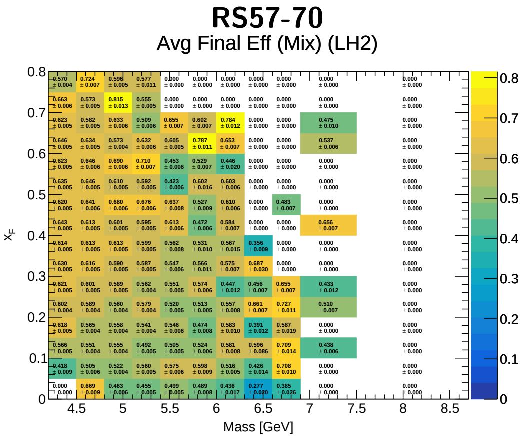


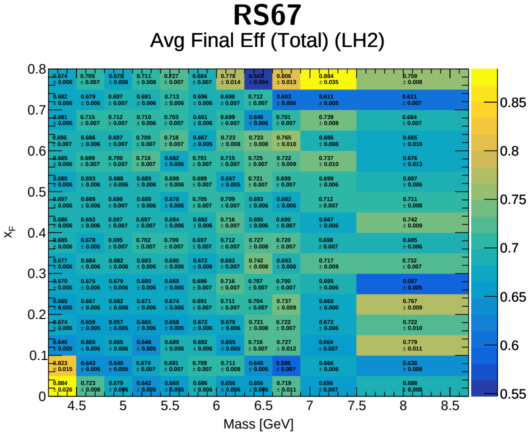
RS57-70
Avg Hodo Eff (Total) (Flask)

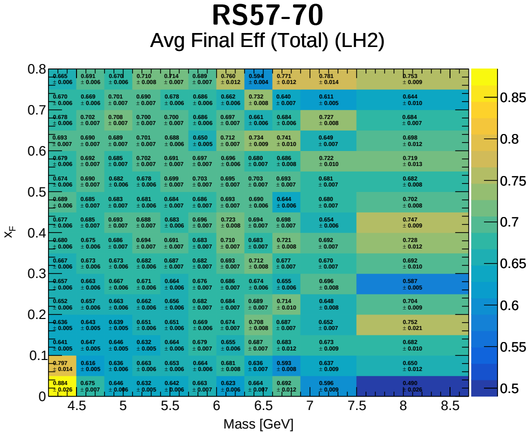


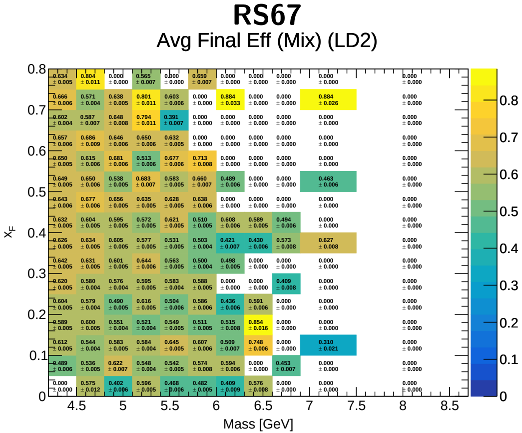
E_mix_final_LH2

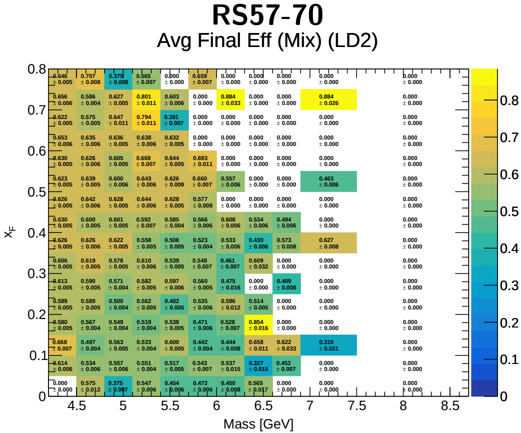




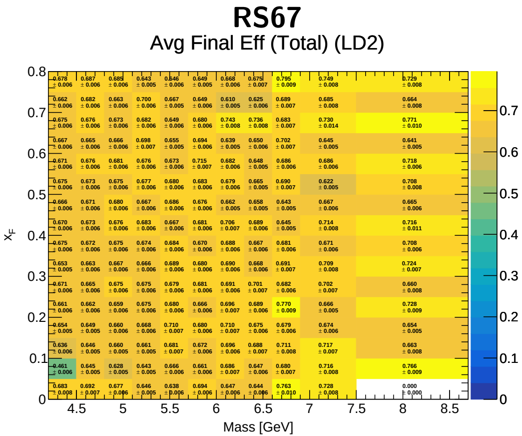

RS67
Avg Final Eff (Total) (LH2)
x_F
Mass [GeV]
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8
4.5 5 5.5 6 6.5 7 7.5 8 8.5
0.55 0.6 0.65 0.7 0.75 0.8 0.85

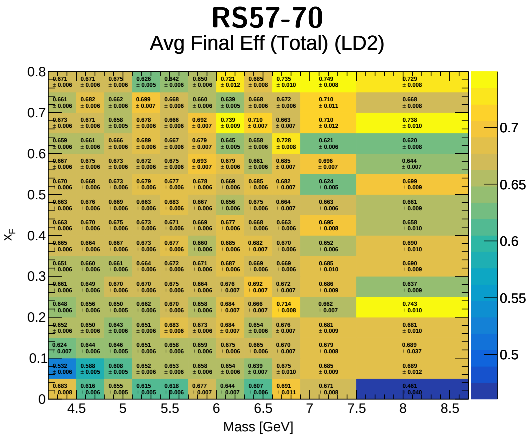

RS57-70
Avg Final Eff (Total) (LH2)
x_F
Mass [GeV]
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8
4.5 5 5.5 6 6.5 7 7.5 8 8.5
0.5 0.55 0.6 0.65 0.7 0.75 0.8 0.85


RS67
Avg Final Eff (Mix) (LD2)
x_F
Mass [GeV]
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
4.5
5
5.5
6
6.5
7
7.5
8
8.5

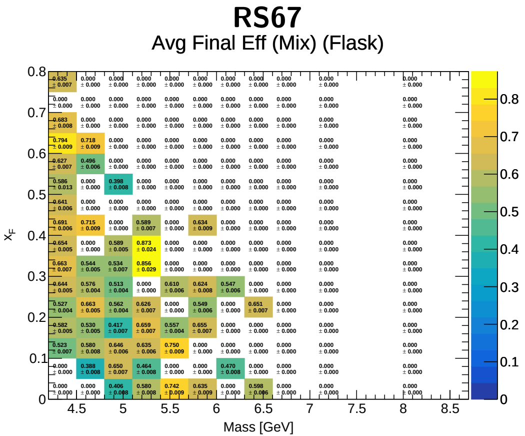

RS57-70
Avg Final Eff (Mix) (LD2)
x_F
Mass [GeV]
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
4.5
5
5.5
6
6.5
7
7.5
8
8.5

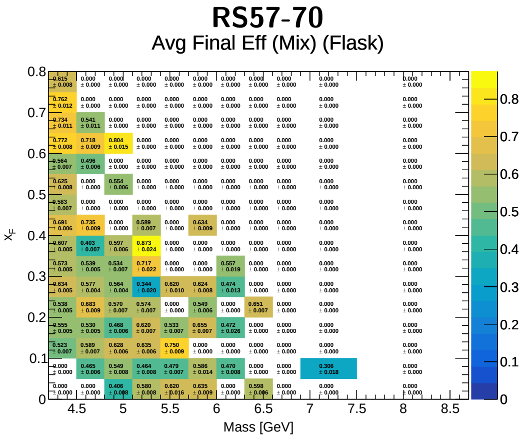
E_total_final_LD2



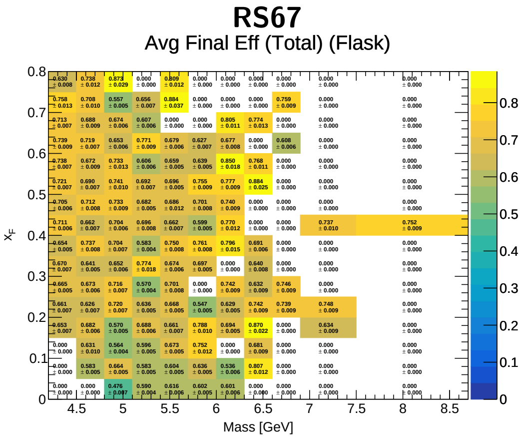


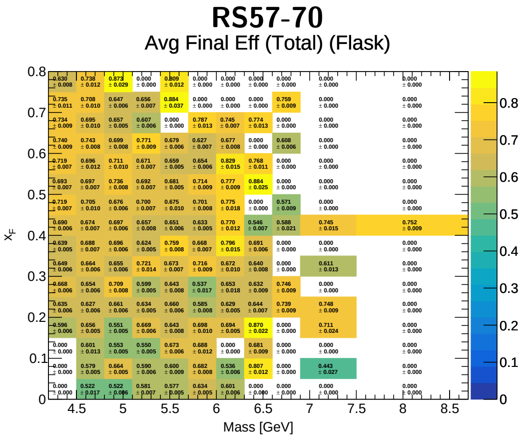
E_mix_final_Flask



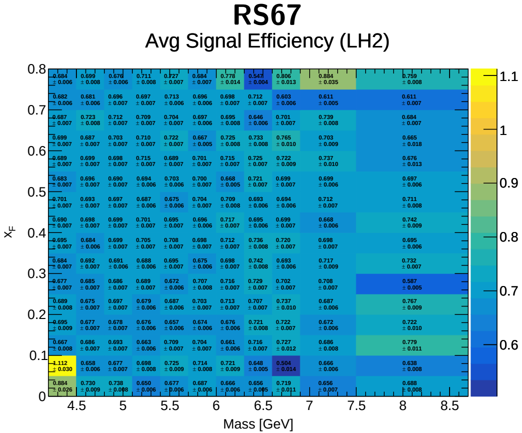


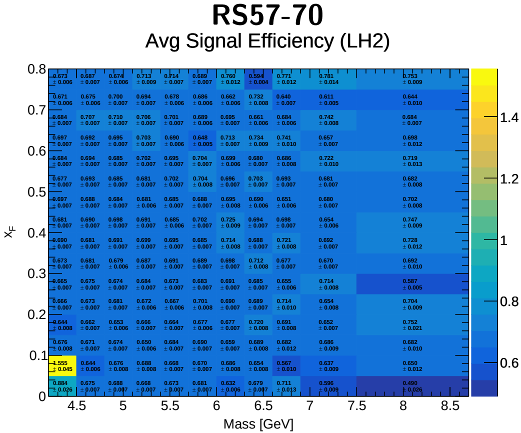
E_total_final_Flask



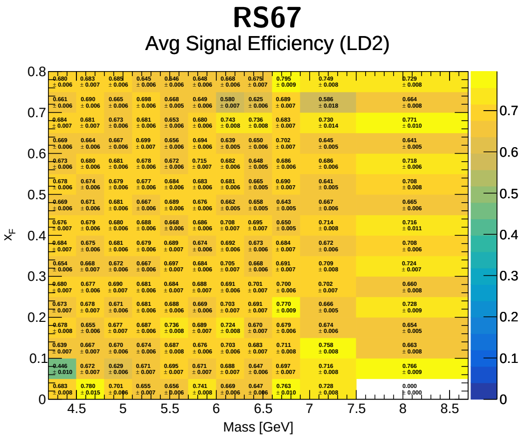


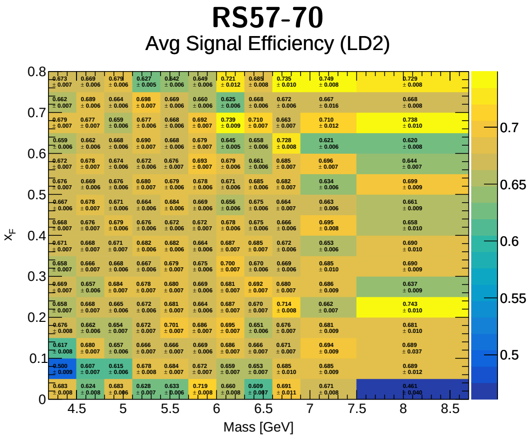
E_final_signal_LH2





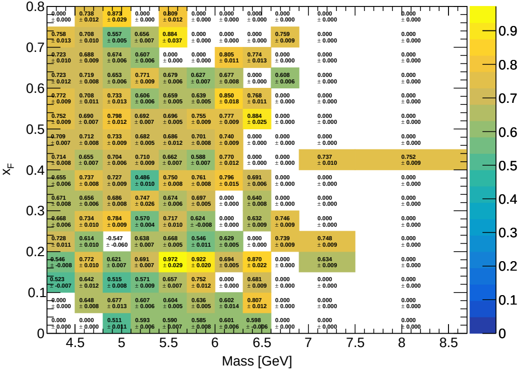
E_final_signal_LD2



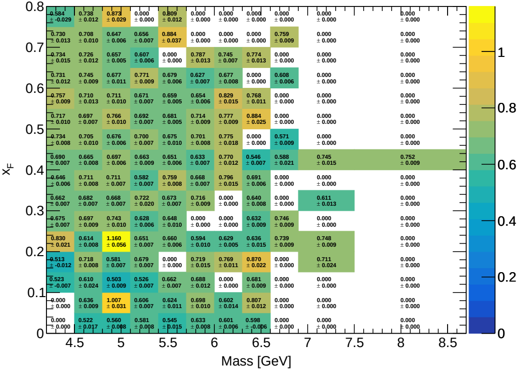


E_final_signal_Flask

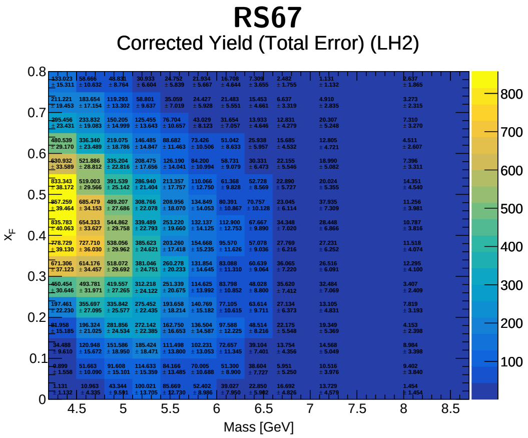
RS67
Avg Signal Efficiency (Flask)

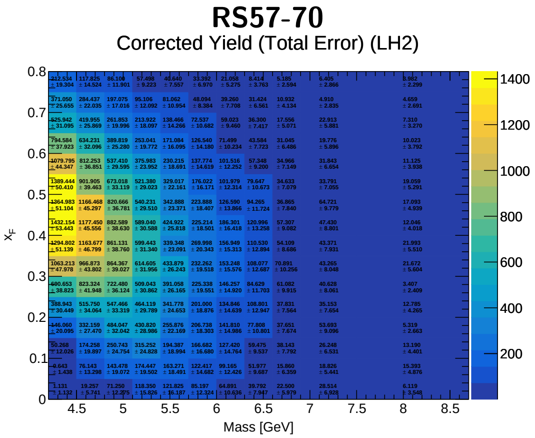


RS57-70
Avg Signal Efficiency (Flask)

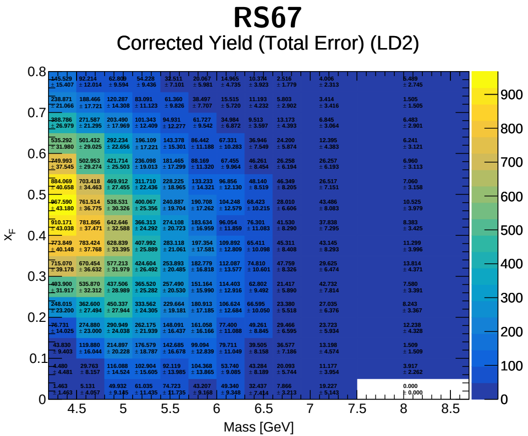


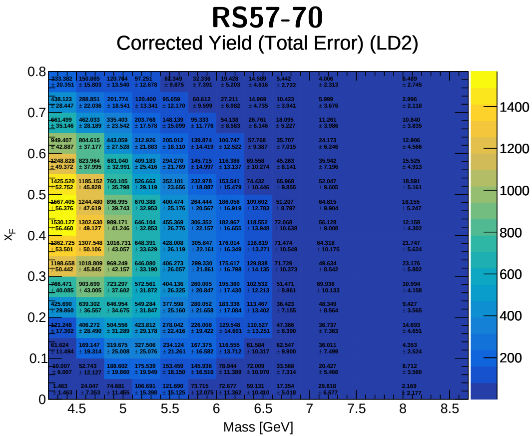
Y_corrected_LH2



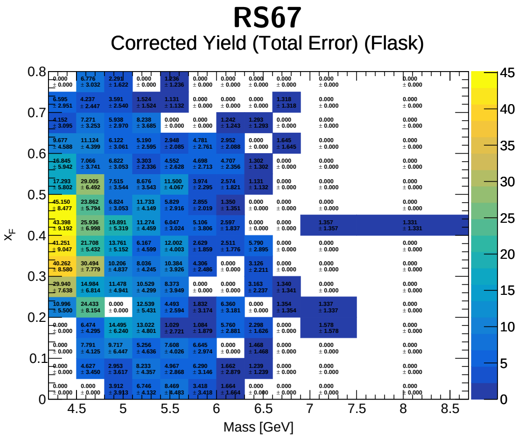


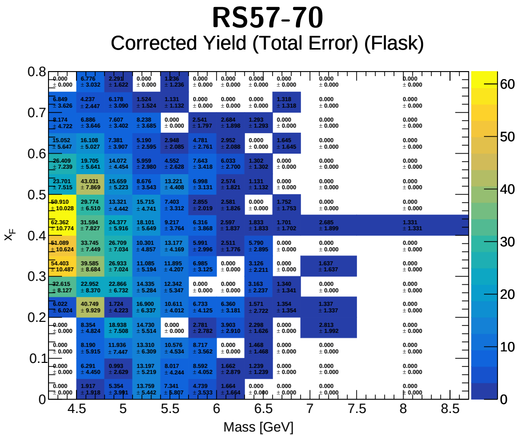
Y_corrected_LD2



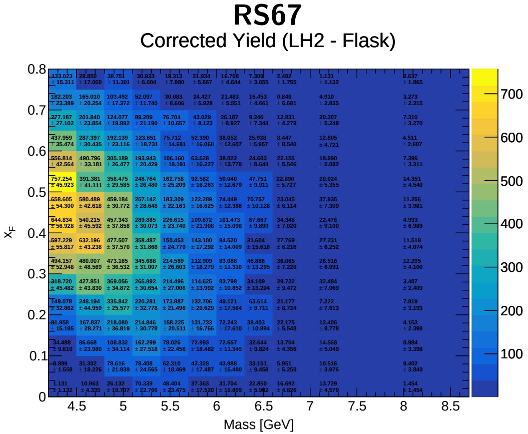


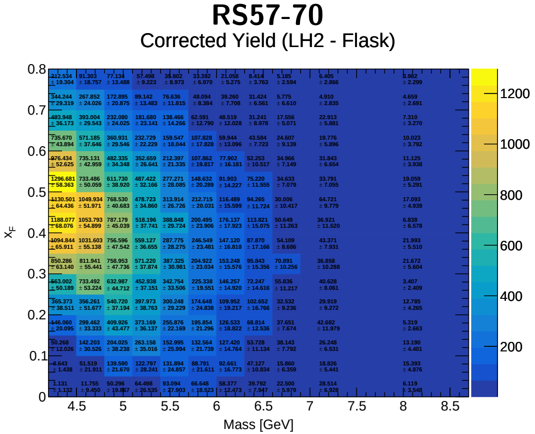
Y_corrected_Flask



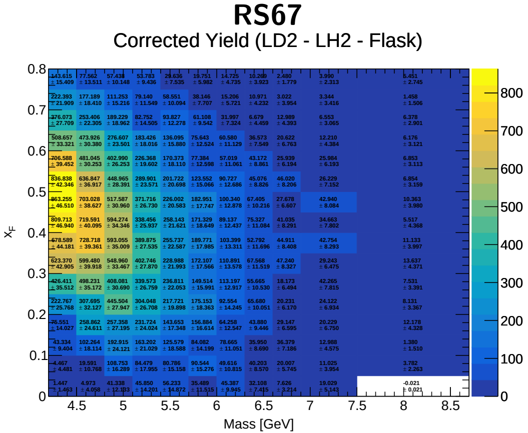


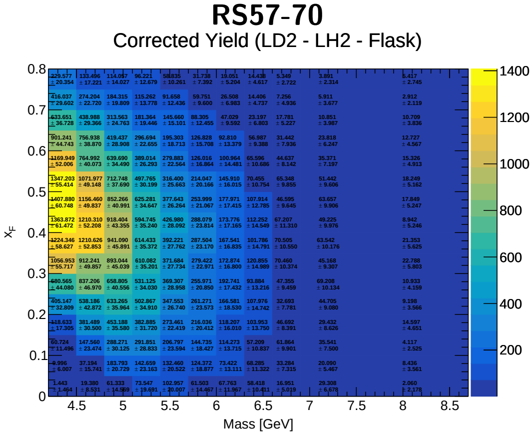
Y_corrected_Subtracted_LH2





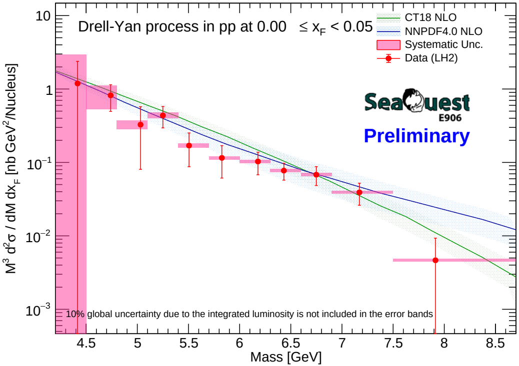
Y_corrected_Subtracted_LD2



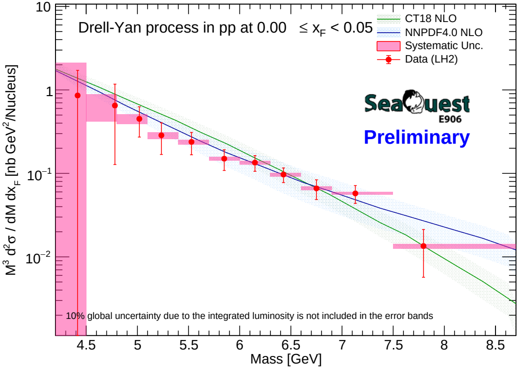


CrossSection LH2 $0.00 \leq xF < 0.05$ (Centroid)

RS67

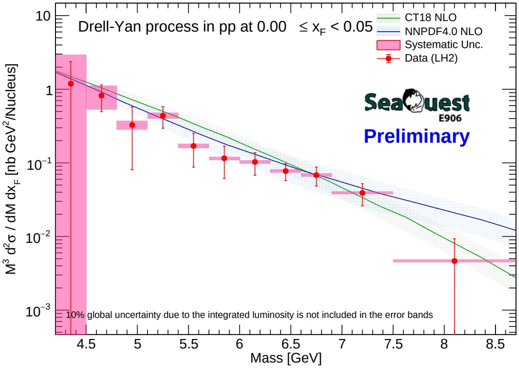


RS57-70



CrossSection LH2 $0.00 \leq xF < 0.05$ (GeoCenter)

RS67



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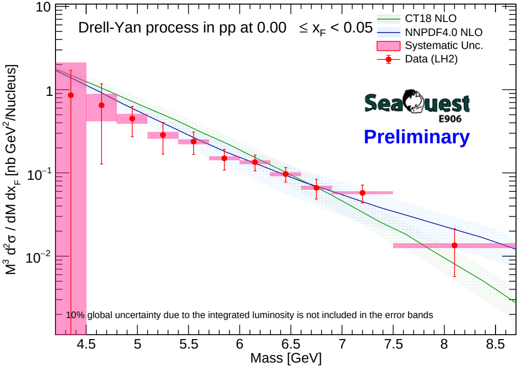
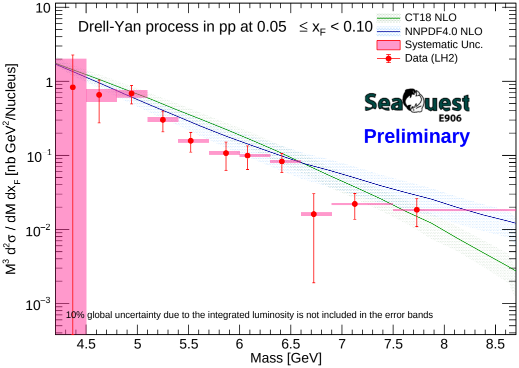


CrossSection LH2 $0.05 \leq$ xF < 0.10 (Centroid)

RS67



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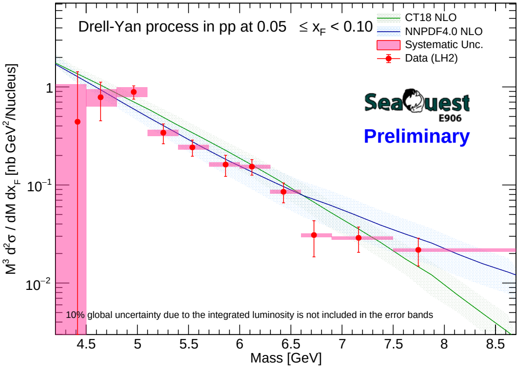
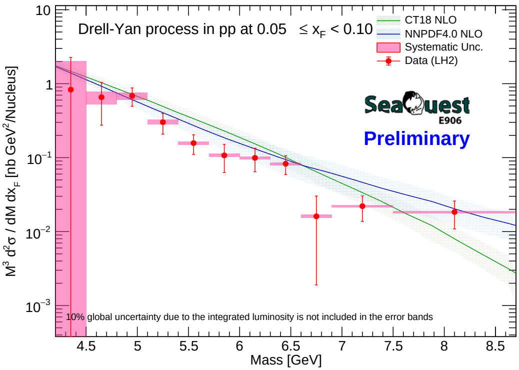


CrossSection LH2 $0.05 \leq xF < 0.10$ (GeoCenter)

RS67



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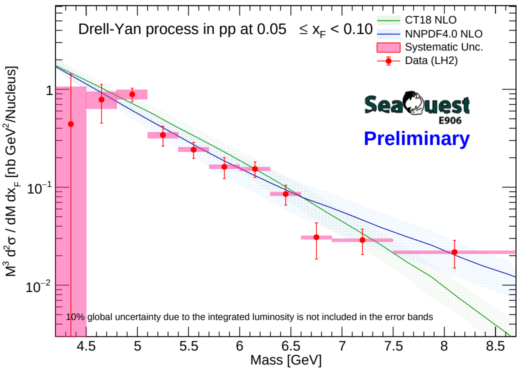
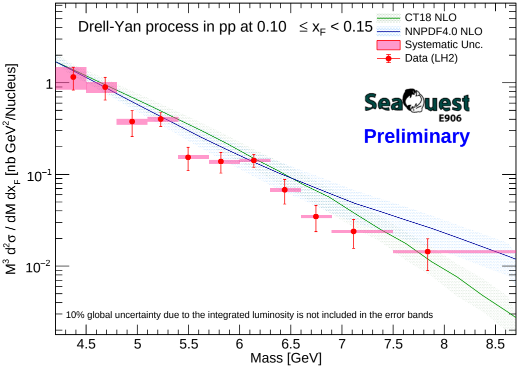


CrossSection LH2 $0.10 \leq xF < 0.15$ (Centroid)

RS67



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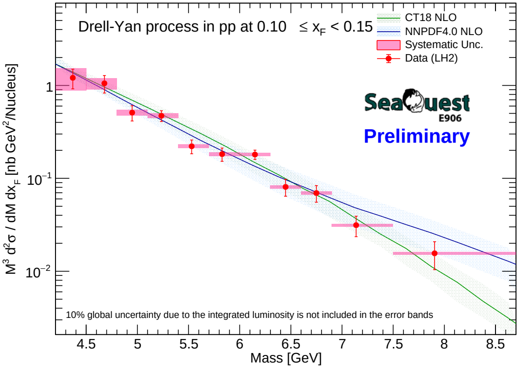
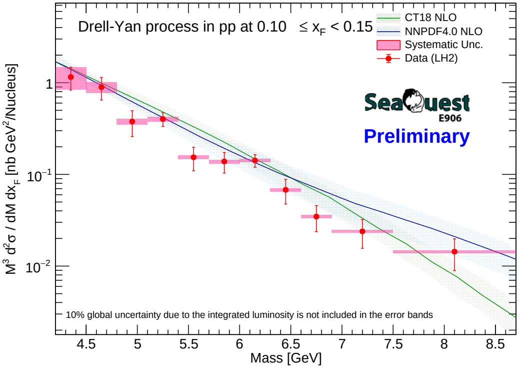


CrossSection LH2 0.10 $\leq$ xF $<$ 0.15 (GeoCenter)

RS67



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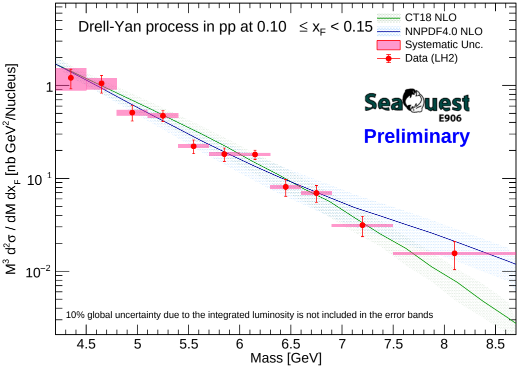
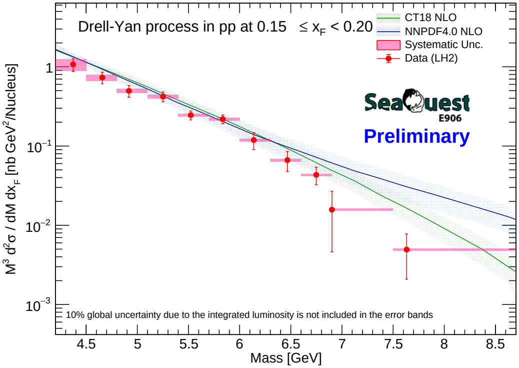
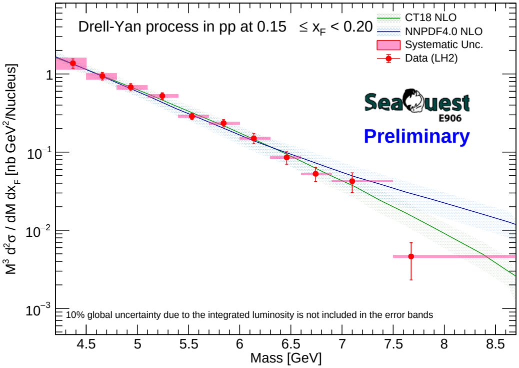


CrossSection LH2 $0.15 \leq xF < 0.20$ (Centroid)

RS67

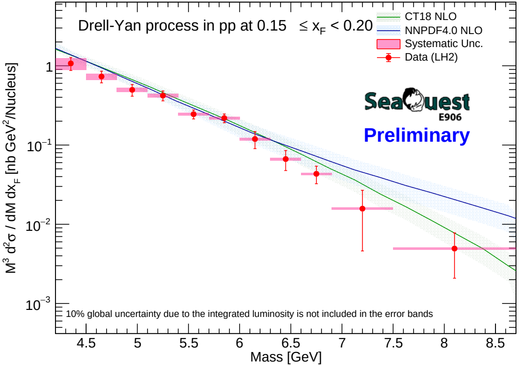


RS57-70



CrossSection LH2 0.15 $\leq$ xF $<$ 0.20 (GeoCenter)

RS67



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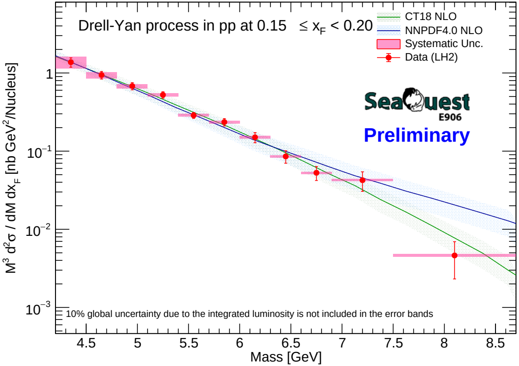
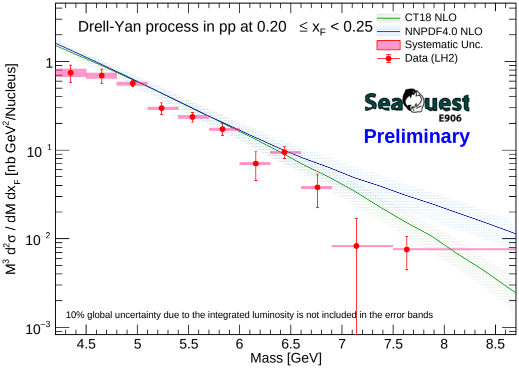


CrossSection LH2 $0.20 \leq xF < 0.25$ (Centroid)

RS67



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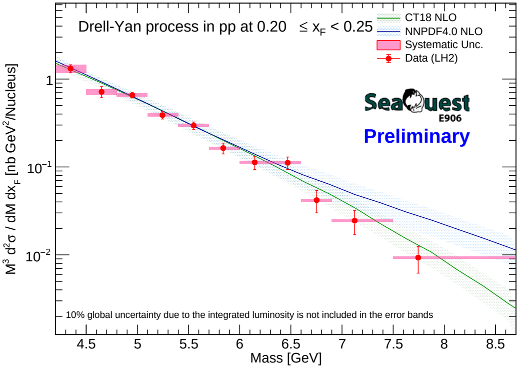
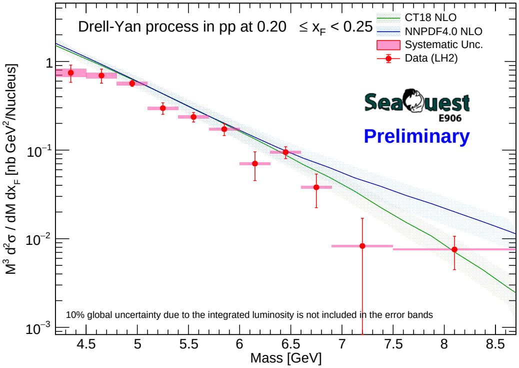


CrossSection LH2 $0.20 \leq xF < 0.25$ (GeoCenter)

RS67



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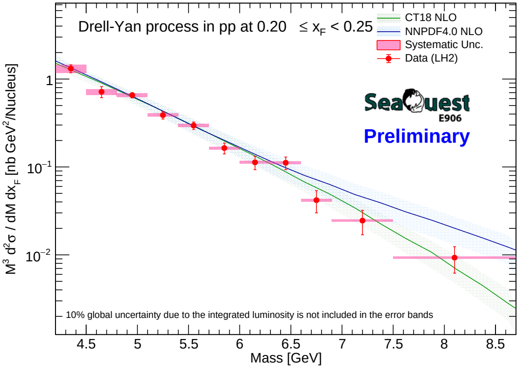
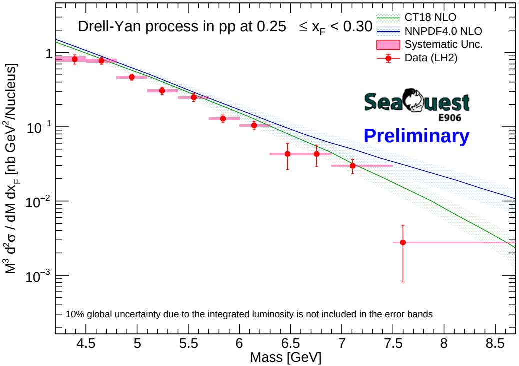


CrossSection LH2 $0.25 \leq xF < 0.30$ (Centroid)

RS67



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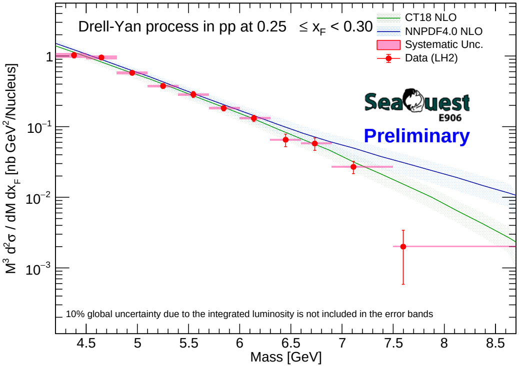
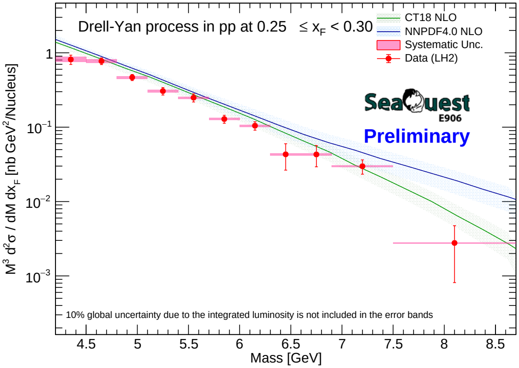


CrossSection LH2 $0.25 \leq xF < 0.30$ (GeoCenter)

RS67



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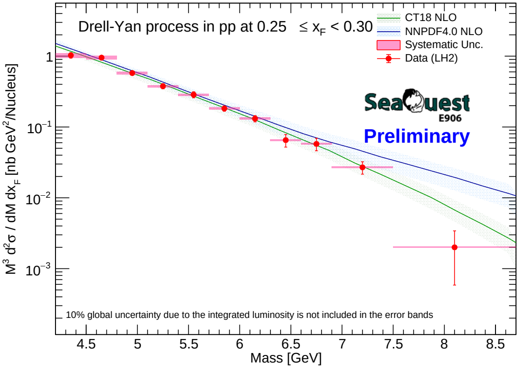
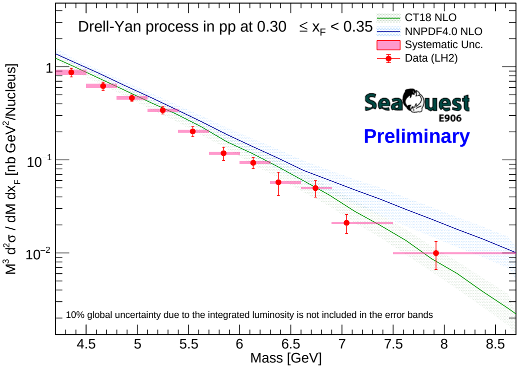


CrossSection LH2 $0.30 \leq xF < 0.35$ (Centroid)

RS67



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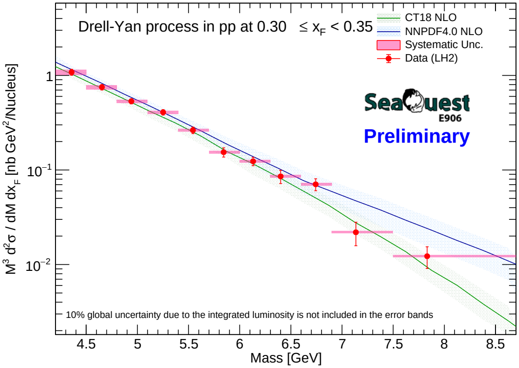
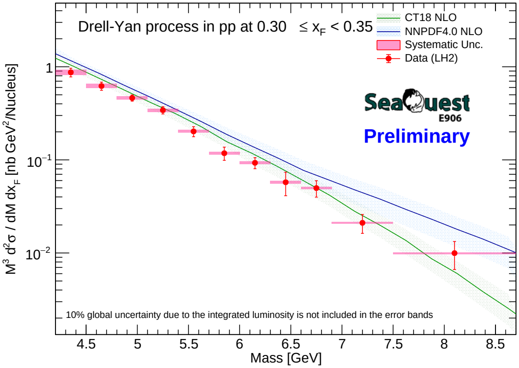


CrossSection LH2 0.30 $\leq$ xF $<$ 0.35 (GeoCenter)

RS67



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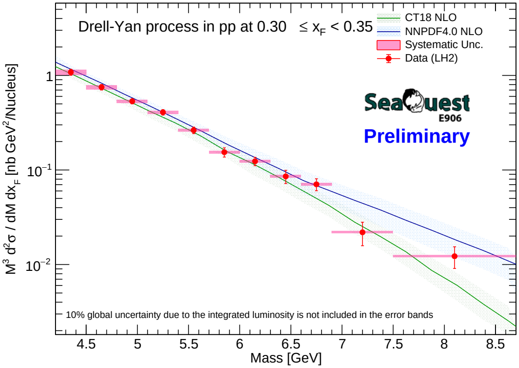
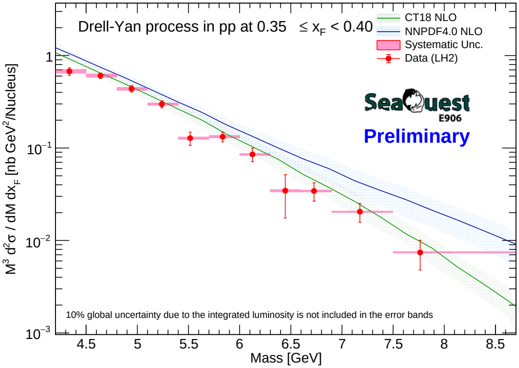


CrossSection LH2 $0.35 \leq xF < 0.40$ (Centroid)

RS67



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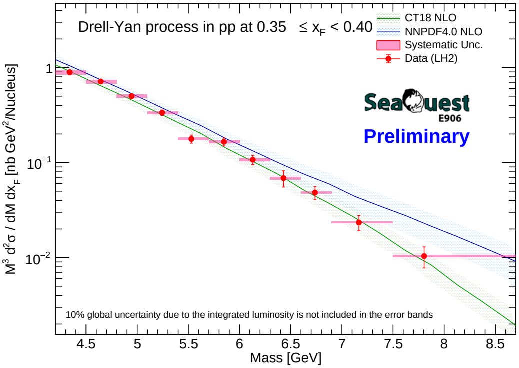
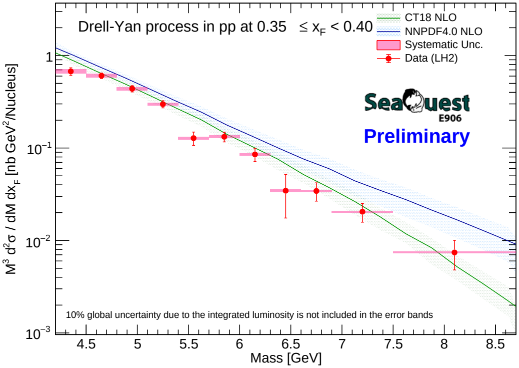


CrossSection LH2 $0.35 \leq xF < 0.40$ (GeoCenter)

RS67



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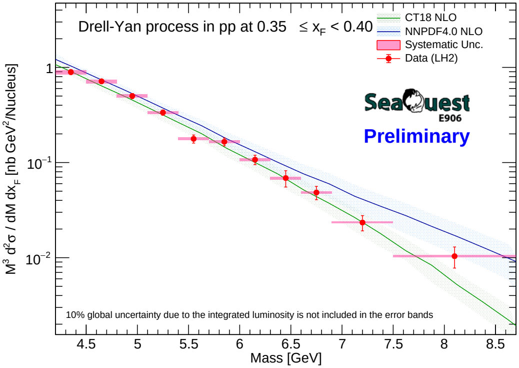
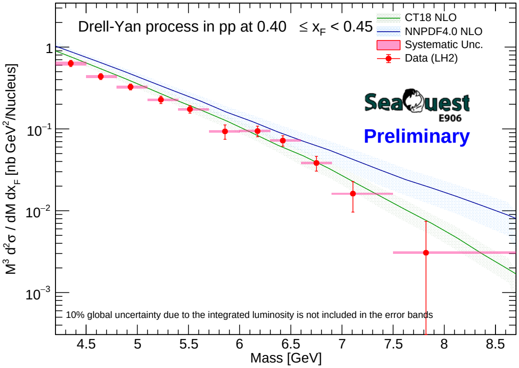


CrossSection LH2 0.40 $\leq$ xF $<$ 0.45 (Centroid)

RS67



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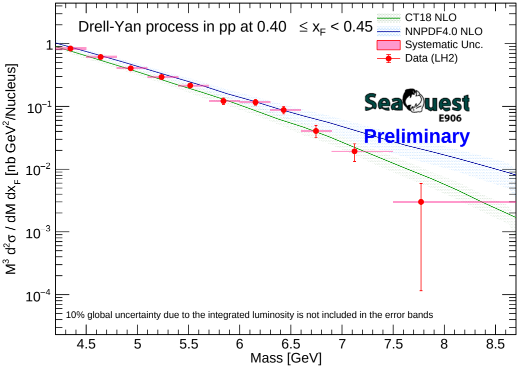
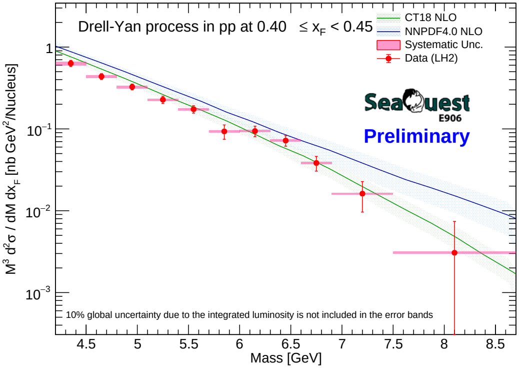


CrossSection LH2 0.40 $\leq$ xF $<$ 0.45 (GeoCenter)

RS67



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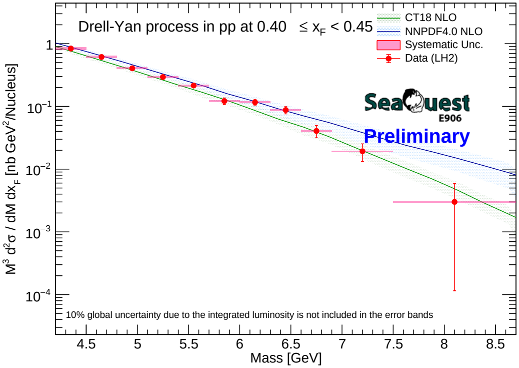
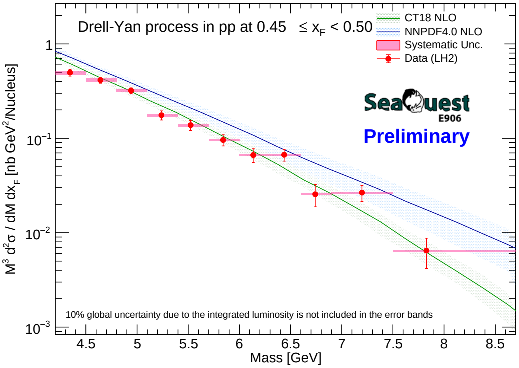


CrossSection LH2 $0.45 \leq xF < 0.50$ (Centroid)

RS67



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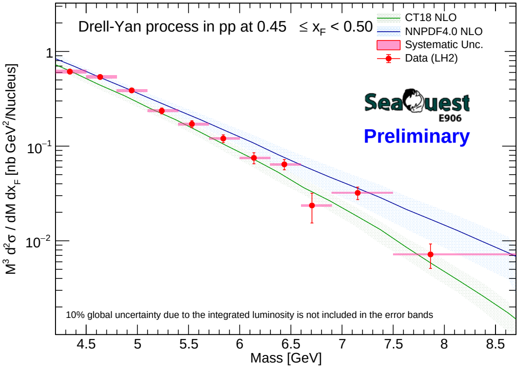
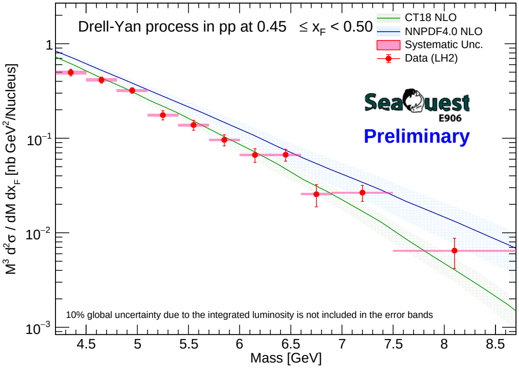
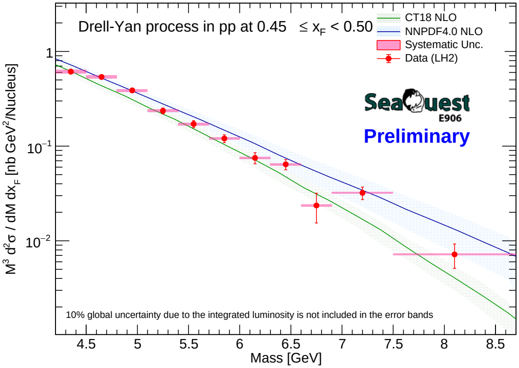


CrossSection LH2 $0.45 \leq xF < 0.50$ (GeoCenter)

RS67

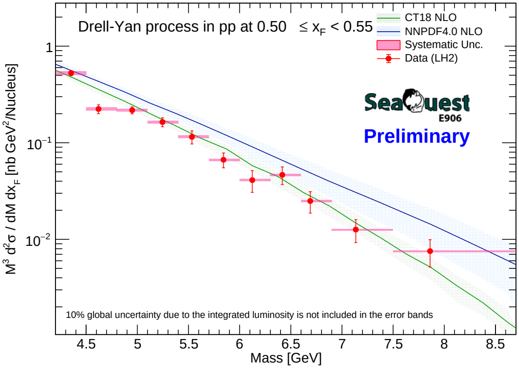


RS57-70



CrossSection LH2 $0.50 \leq xF < 0.55$ (Centroid)

RS67



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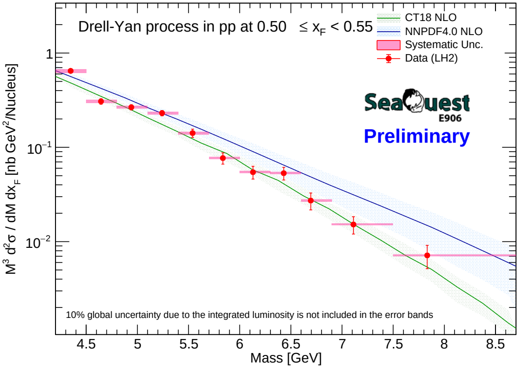
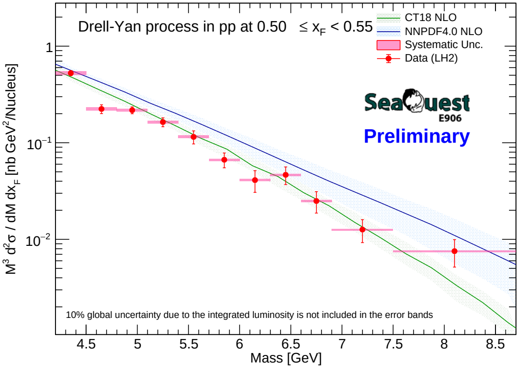


CrossSection LH2 0.50 ≤ xF < 0.55 (GeoCenter)

RS67



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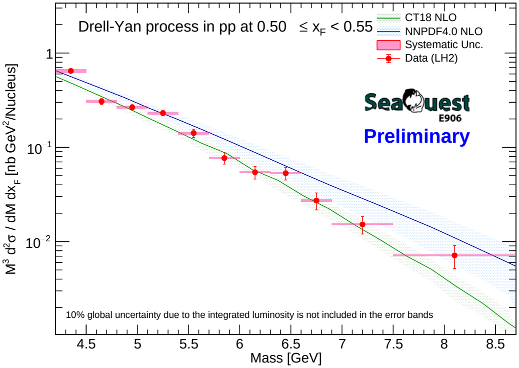
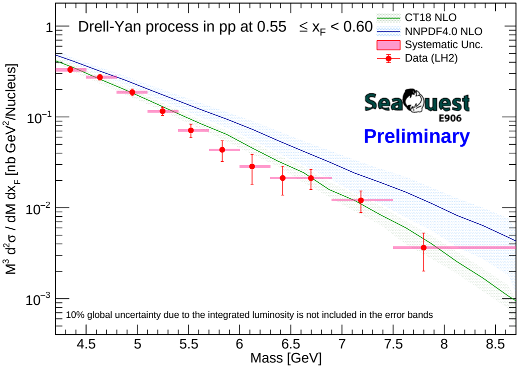


CrossSection LH2 $0.55 \leq xF < 0.60$ (Centroid)

RS67



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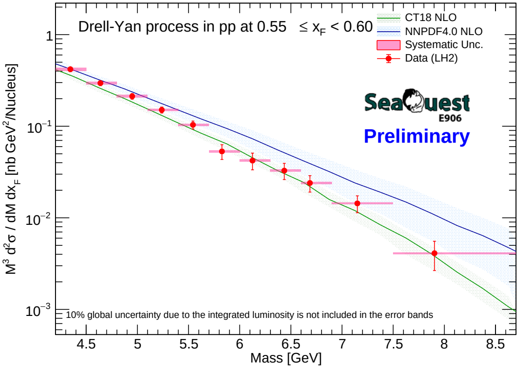
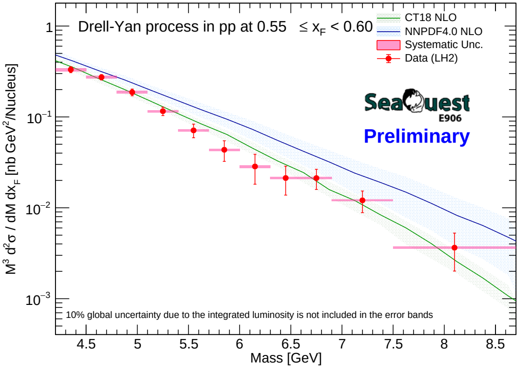


CrossSection LH2 0.55 $\leq$ xF $<$ 0.60 (GeoCenter)

RS67



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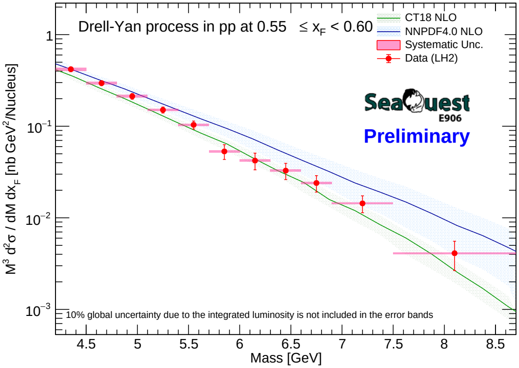
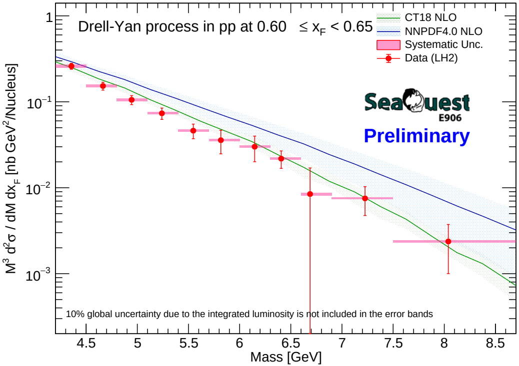


CrossSection LH2 $0.60 \leq xF < 0.65$ (Centroid)

RS67



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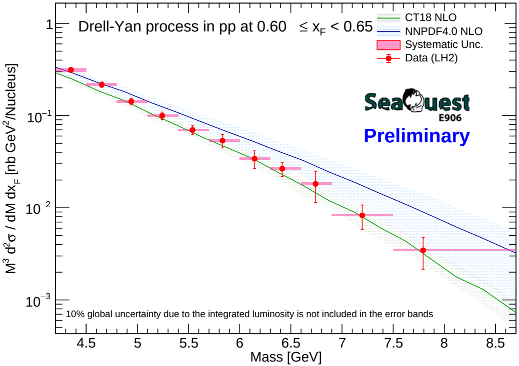
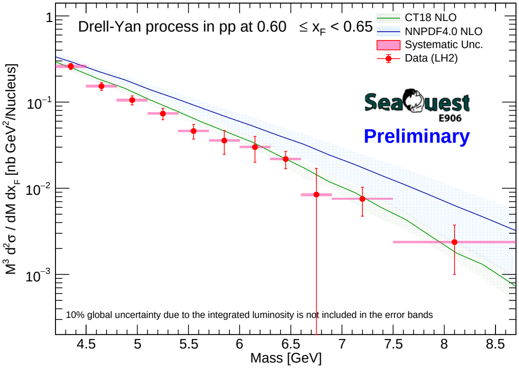


CrossSection LH2 0.60 $\leq$ xF $<$ 0.65 (GeoCenter)

RS67



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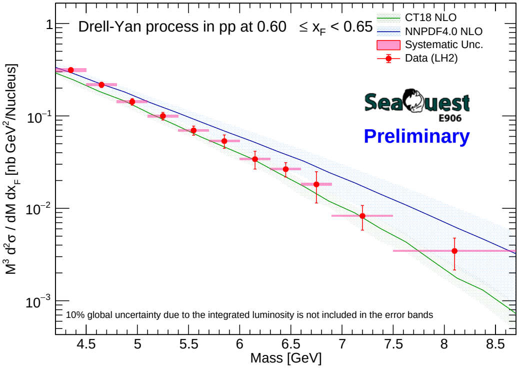
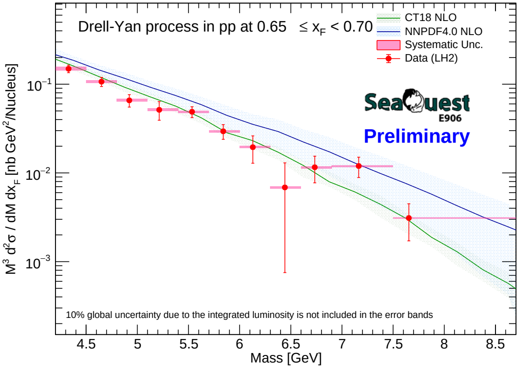
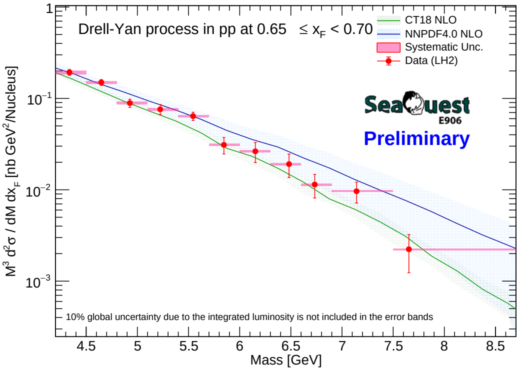


CrossSection LH2 0.65 ≤ xF < 0.70 (Centroid)

RS67

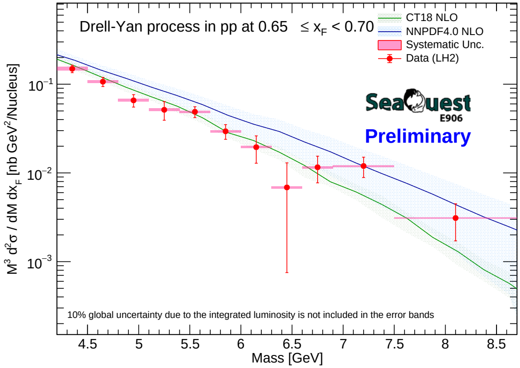


RS57-70



CrossSection LH2 0.65 $\leq$ xF $<$ 0.70 (GeoCenter)

RS67



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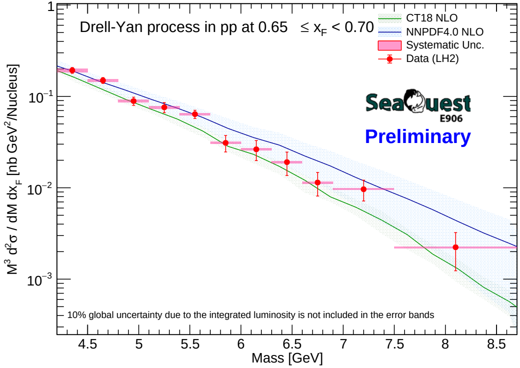
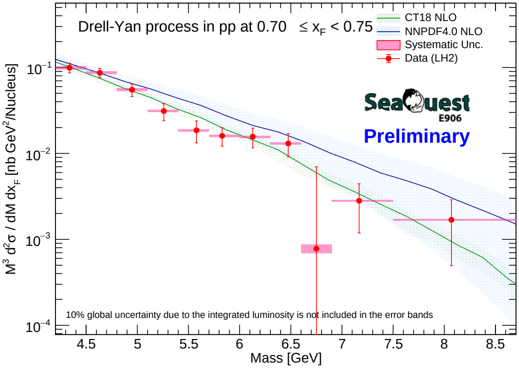


CrossSection LH2 $0.70 \leq xF < 0.75$ (Centroid)

RS67



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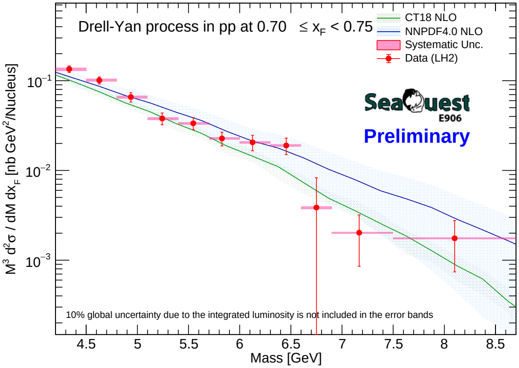
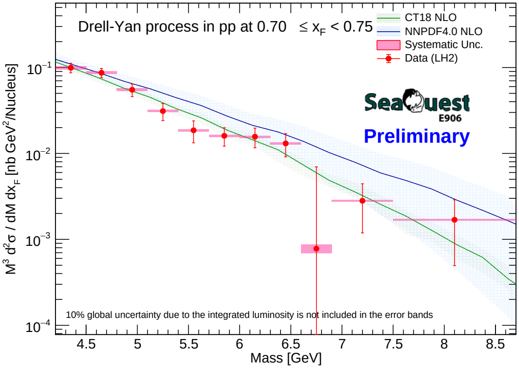
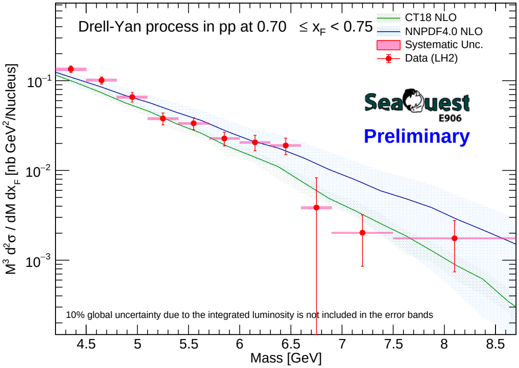


RS67

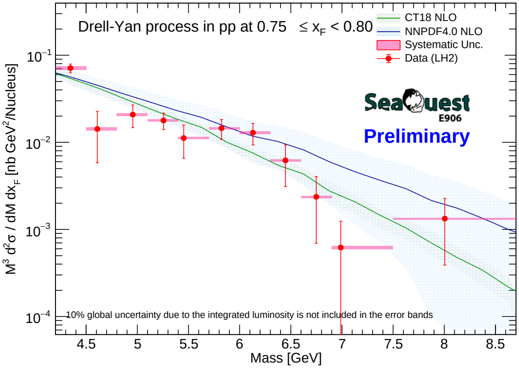

Drell-Yan process in pp at 0.70 ≤ x_F < 0.75
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

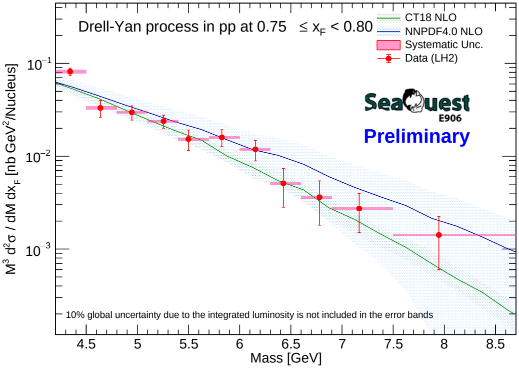

Drell-Yan process in pp at 0.70 ≤ x_F < 0.75
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection LH2 $0.75 \leq xF < 0.80$ (Centroid)

RS67

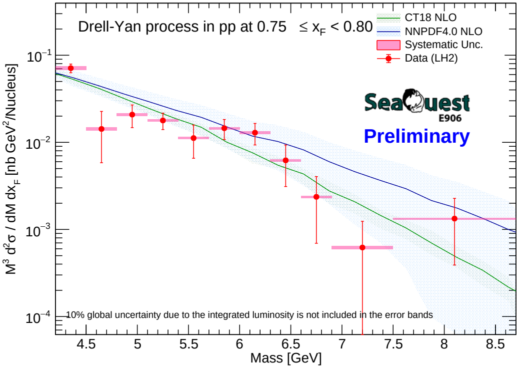


RS57-70

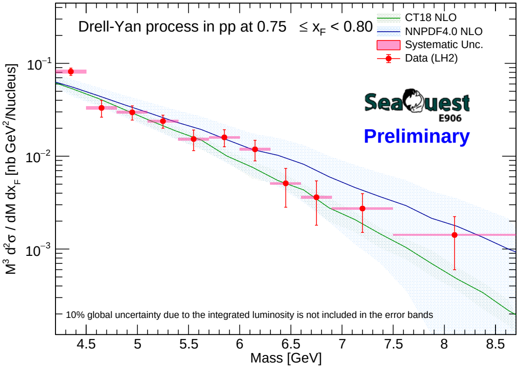


CrossSection LH2 $0.75 \leq xF < 0.80$ (GeoCenter)

RS67

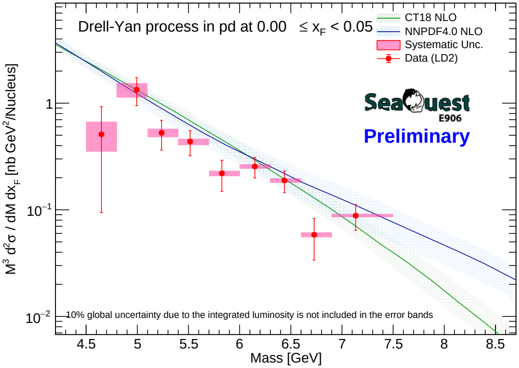


RS57-70

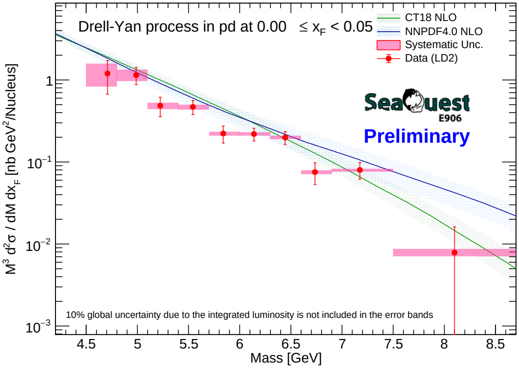


CrossSection LD2 $0.00 \leq xF < 0.05$ (Centroid)

RS67

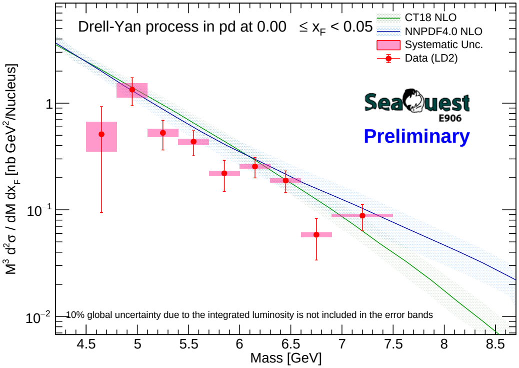


RS57-70

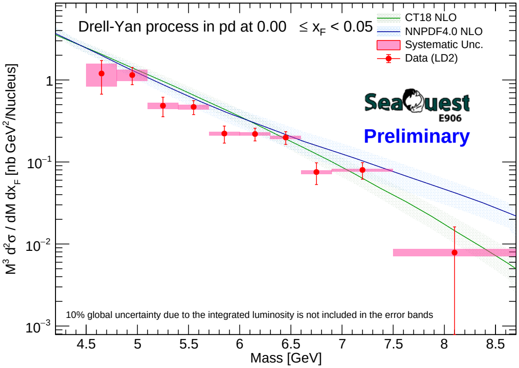


CrossSection LD2 $0.00 \leq xF < 0.05$ (GeoCenter)

RS67

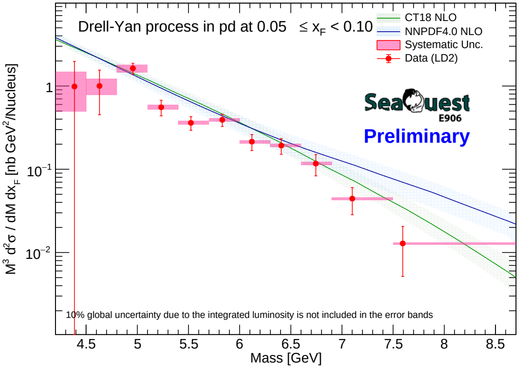


RS57-70

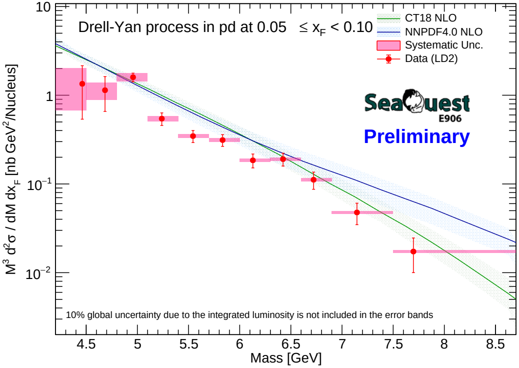


CrossSection LD2 $0.05 \leq xF < 0.10$ (Centroid)

RS67

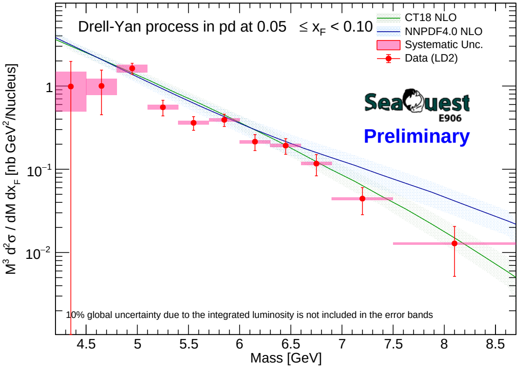


RS57-70

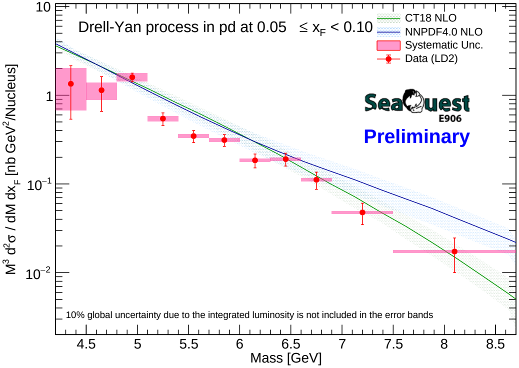


CrossSection LD2 $0.05 \leq xF < 0.10$ (GeoCenter)

RS67

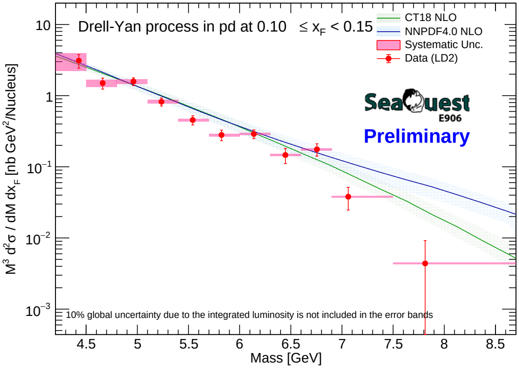


RS57-70

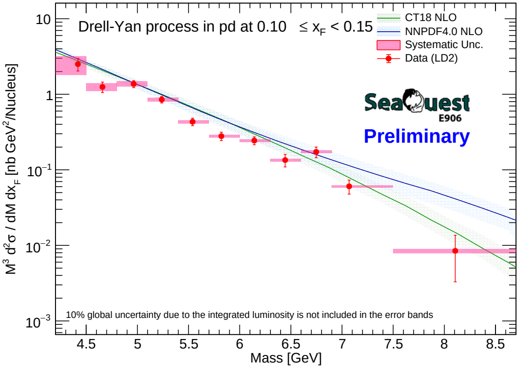


CrossSection LD2 $0.10 \leq xF < 0.15$ (Centroid)

RS67

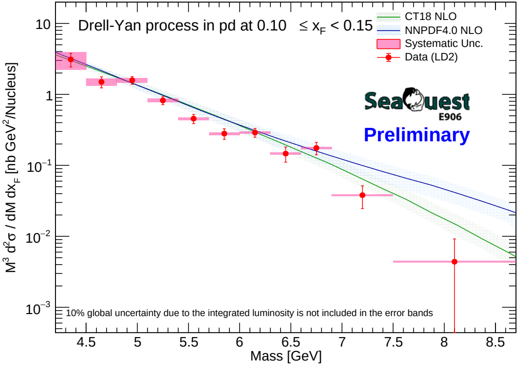


RS57-70

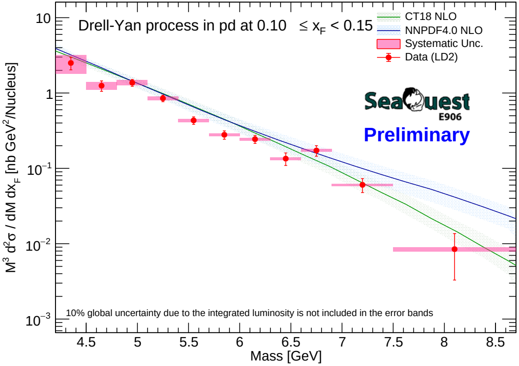


CrossSection LD2 $0.10 \leq xF < 0.15$ (GeoCenter)

RS67

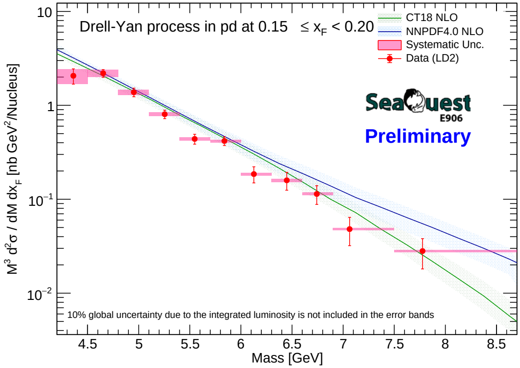


RS57-70

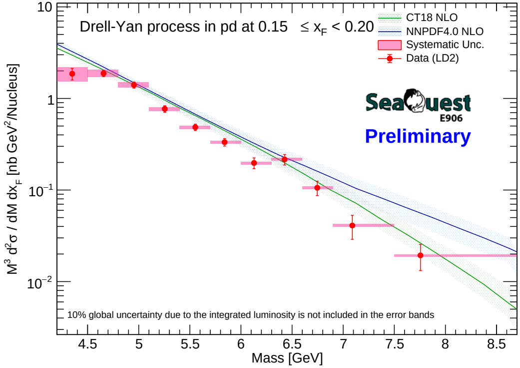


CrossSection LD2 0.15 $\leq$ xF < 0.20 (Centroid)

RS67

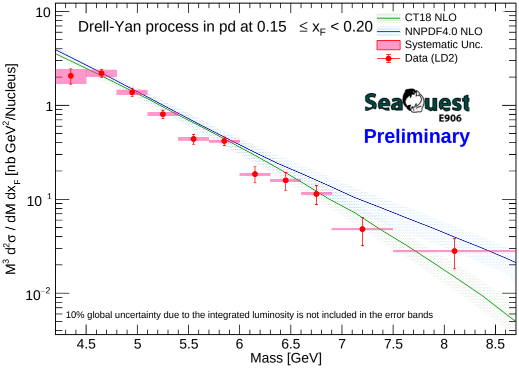


RS57-70

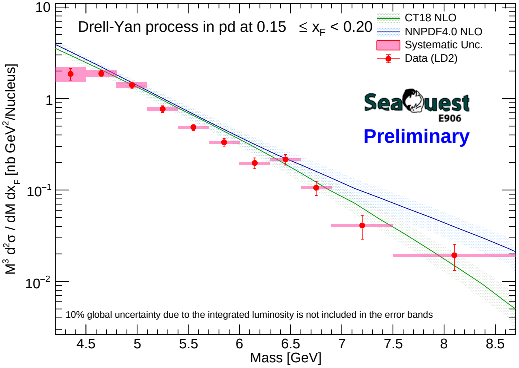


CrossSection LD2 $0.15 \leq xF < 0.20$ (GeoCenter)

RS67

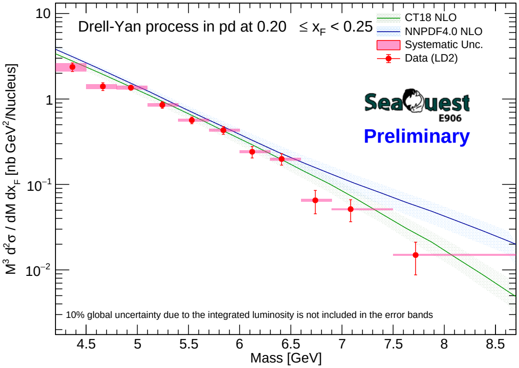


RS57-70

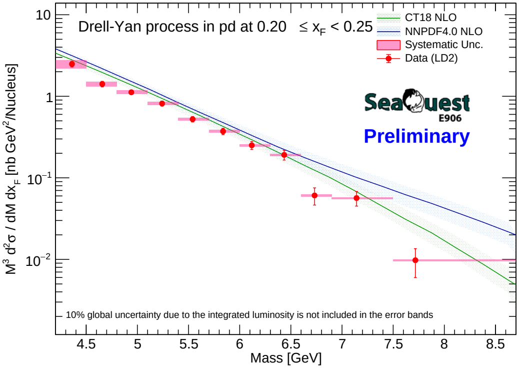


CrossSection LD2 $0.20 \leq xF < 0.25$ (Centroid)

RS67

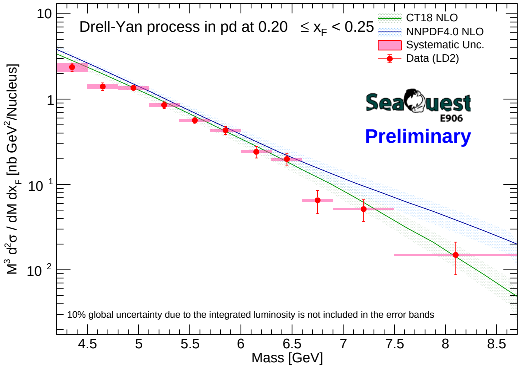


RS57-70

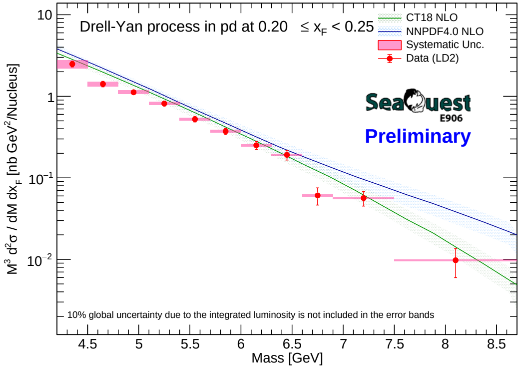


CrossSection LD2 $0.20 \leq xF < 0.25$ (GeoCenter)

RS67

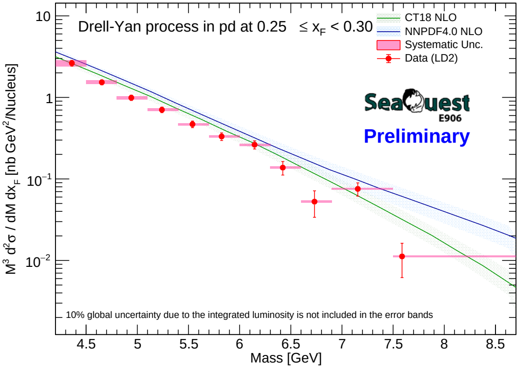


RS57-70

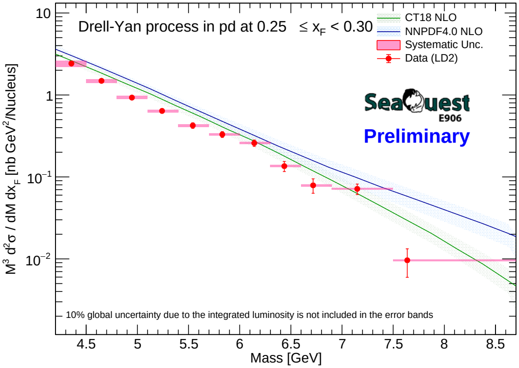


CrossSection LD2 0.25 ≤ xF < 0.30 (Centroid)

RS67

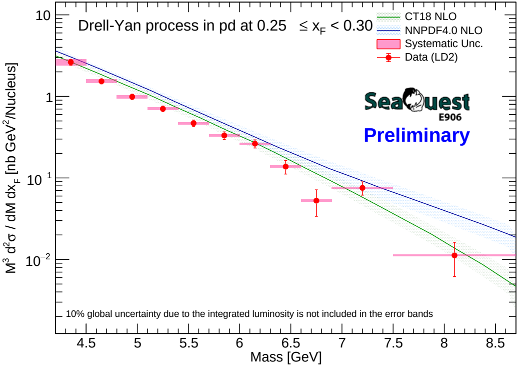


RS57-70

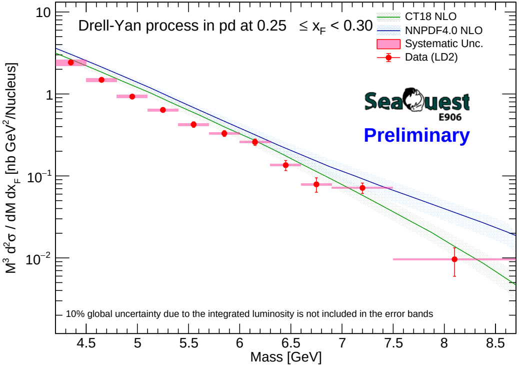


CrossSection LD2 0.25 ≤ xF < 0.30 (GeoCenter)

RS67

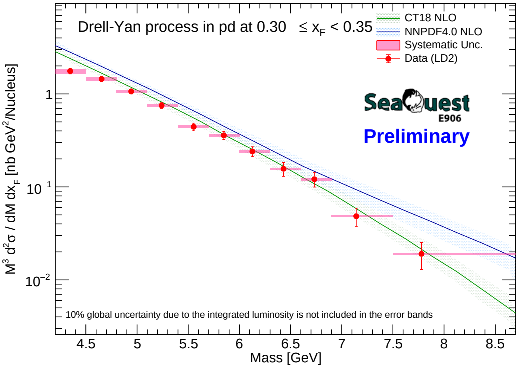


RS57-70

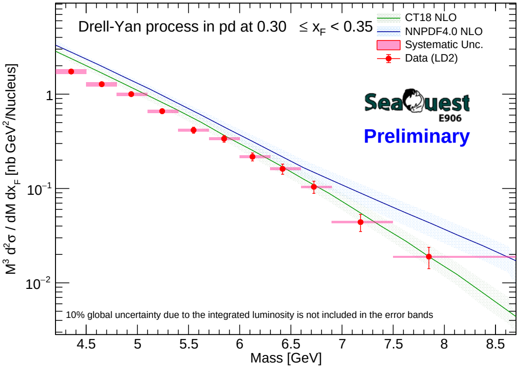


CrossSection LD2 $0.30 \leq xF < 0.35$ (Centroid)

RS67

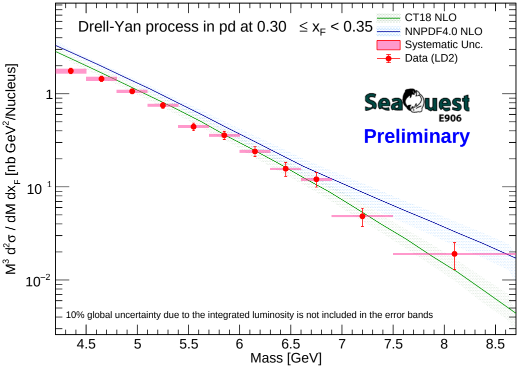


RS57-70

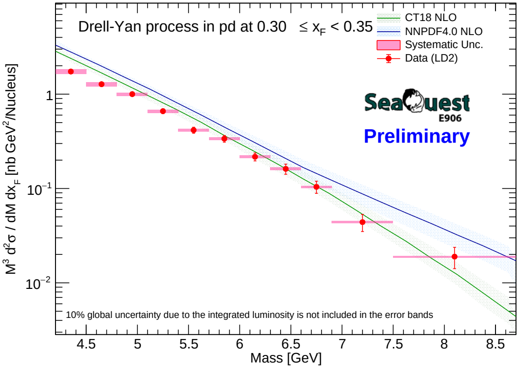


CrossSection LD2 0.30 ≤ xF < 0.35 (GeoCenter)

RS67

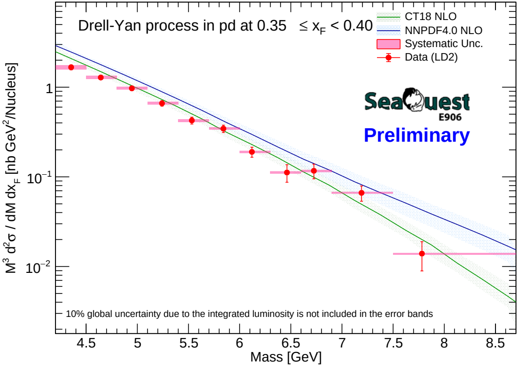


RS57-70

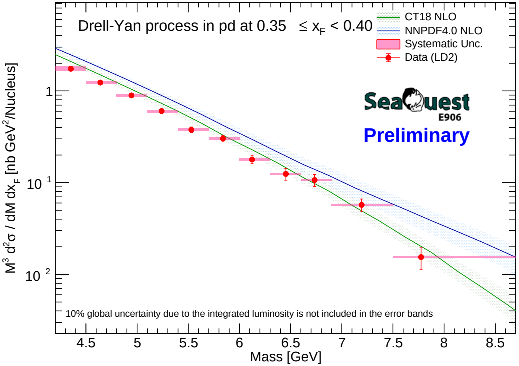


CrossSection LD2 $0.35 \leq xF < 0.40$ (Centroid)

RS67

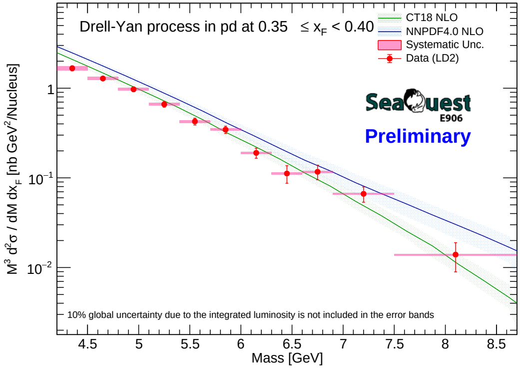


RS57-70

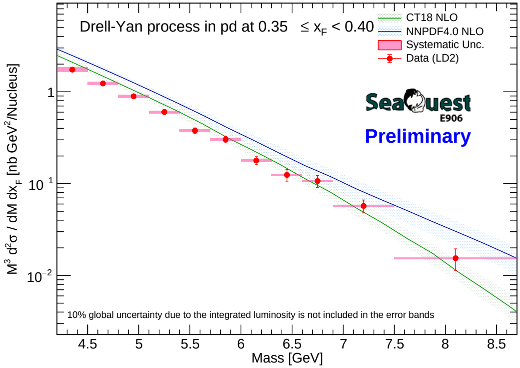


CrossSection LD2 $0.35 \leq xF < 0.40$ (GeoCenter)

RS67

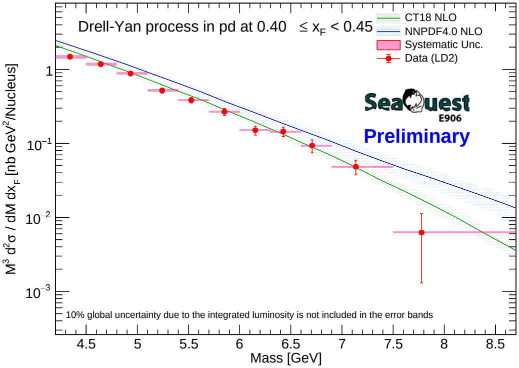


RS57-70

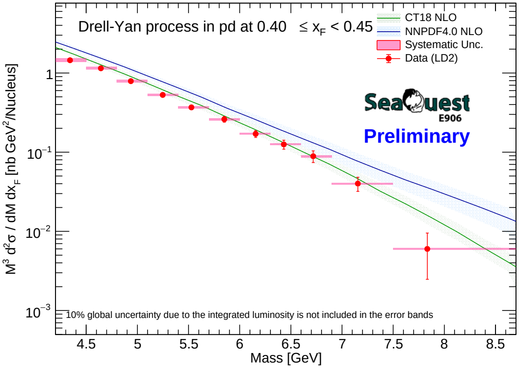


CrossSection LD2 $0.40 \leq xF < 0.45$ (Centroid)

RS67

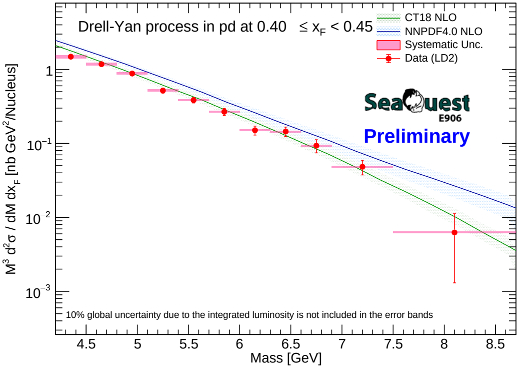


RS57-70

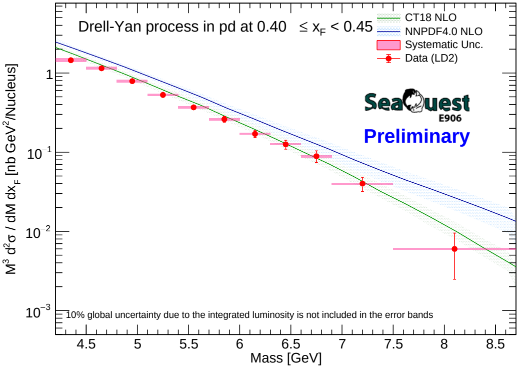


CrossSection LD2 $0.40 \leq xF < 0.45$ (GeoCenter)

RS67

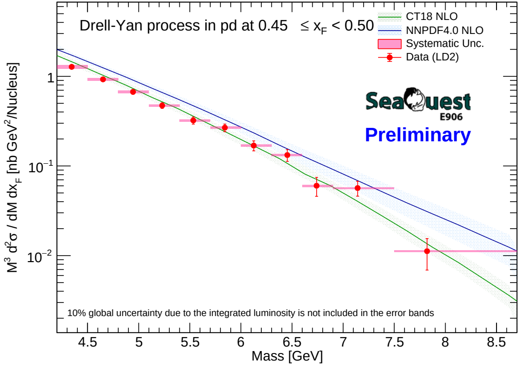


RS57-70

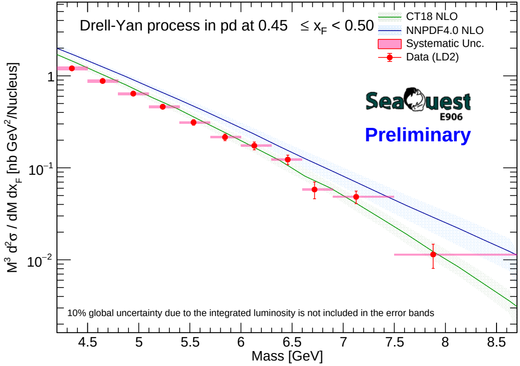


CrossSection LD2 $0.45 \leq xF < 0.50$ (Centroid)

RS67

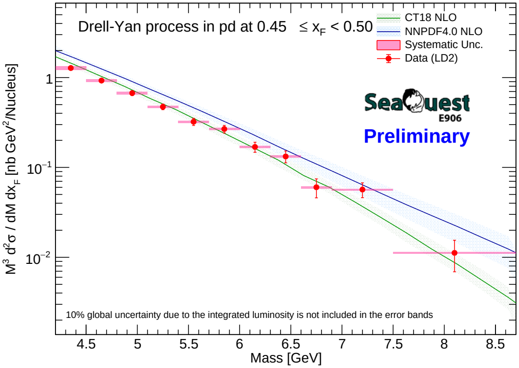


RS57-70

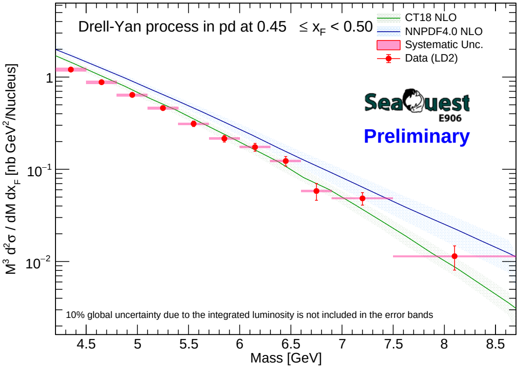


CrossSection LD2 0.45 $\leq$ xF $<$ 0.50 (GeoCenter)

RS67

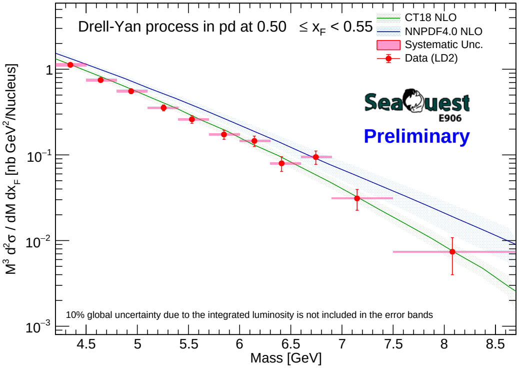


RS57-70

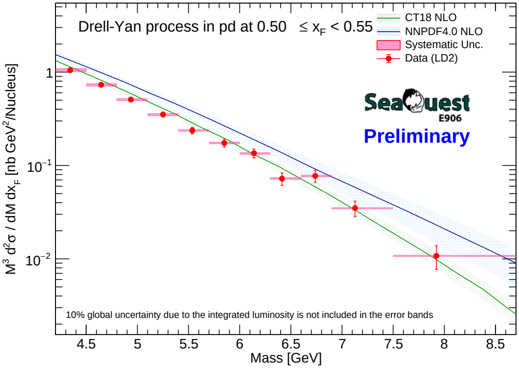


CrossSection LD2 $0.50 \leq xF < 0.55$ (Centroid)

RS67

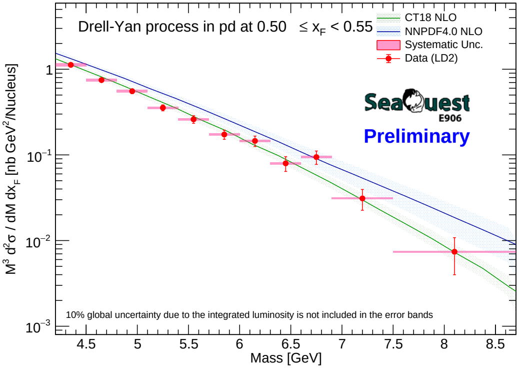


RS57-70

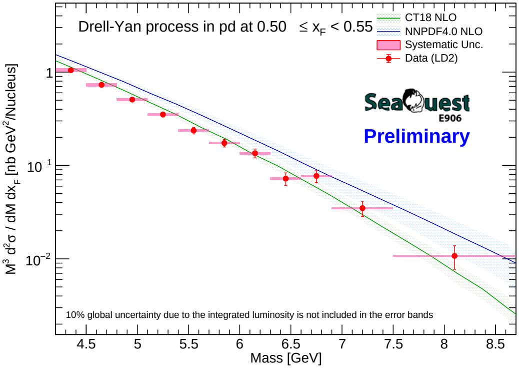


CrossSection LD2 $0.50 \leq xF < 0.55$ (GeoCenter)

RS67

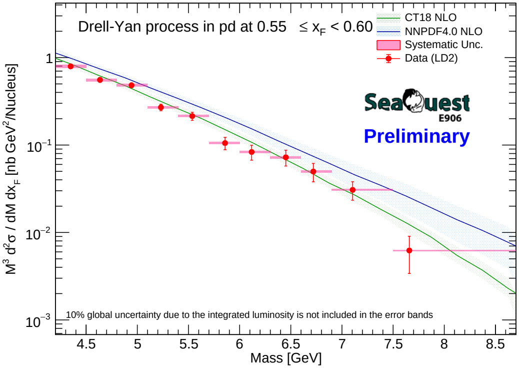


RS57-70

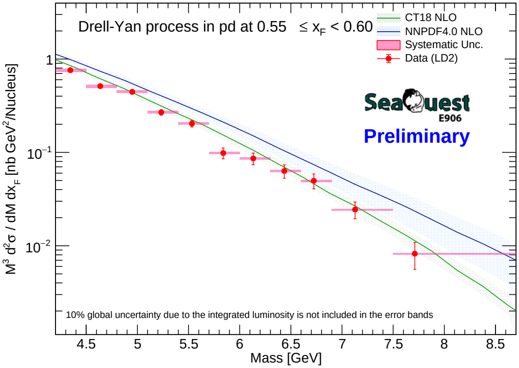


CrossSection LD2 $0.55 \leq xF < 0.60$ (Centroid)

RS67

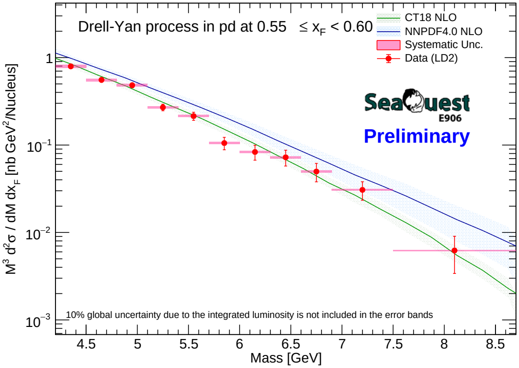


RS57-70

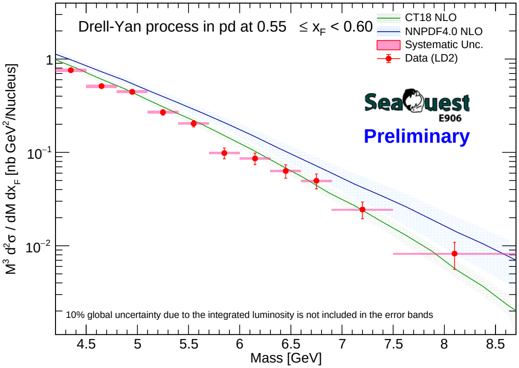


CrossSection LD2 0.55 $\leq$ xF $<$ 0.60 (GeoCenter)

RS67

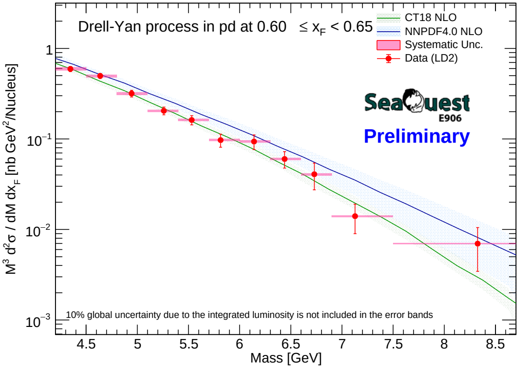


RS57-70



CrossSection LD2 $0.60 \leq xF < 0.65$ (Centroid)

RS67



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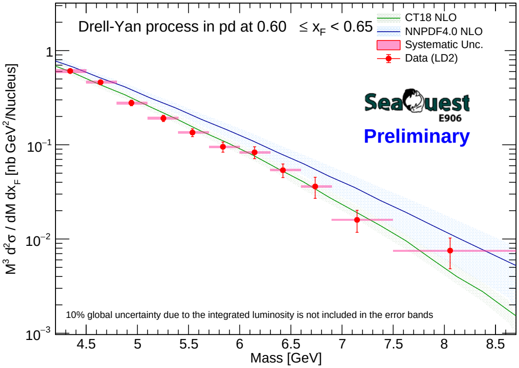
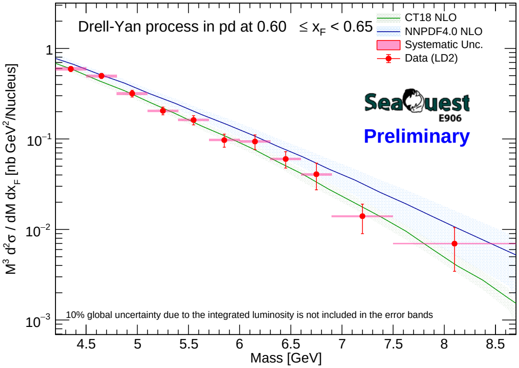


CrossSection LD2 $0.60 \leq xF < 0.65$ (GeoCenter)

RS67



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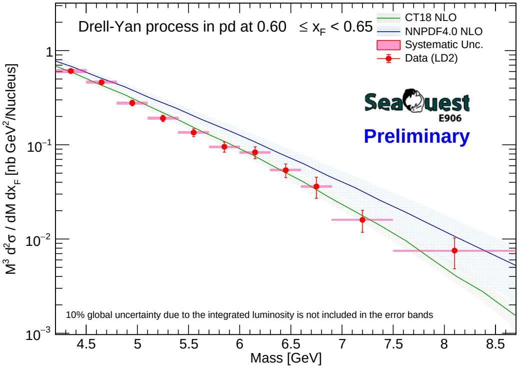
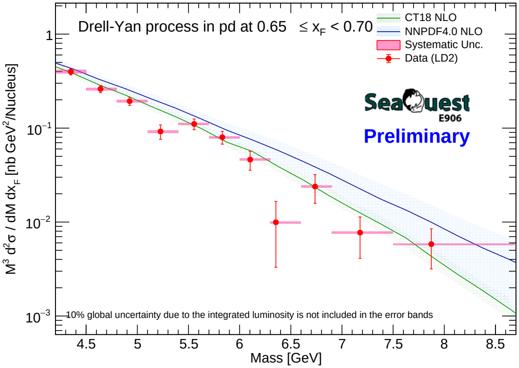


CrossSection LD2 $0.65 \leq xF < 0.70$ (Centroid)

RS67



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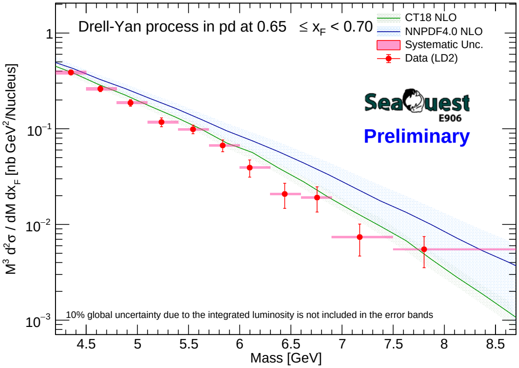
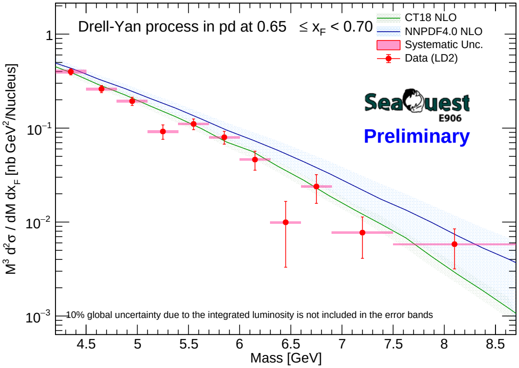


CrossSection LD2 0.65 $\leq$ xF $<$ 0.70 (GeoCenter)

RS67



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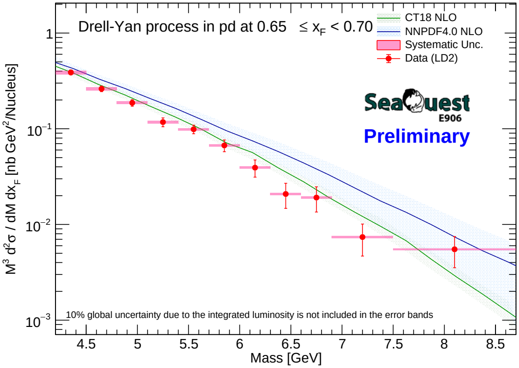
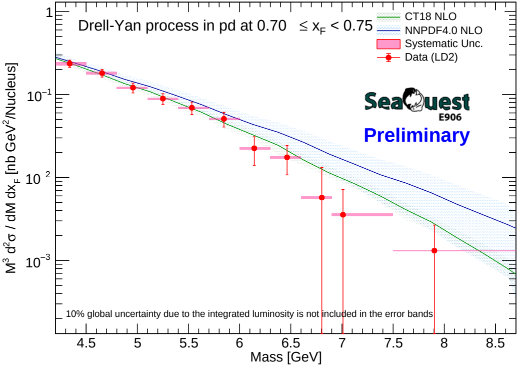
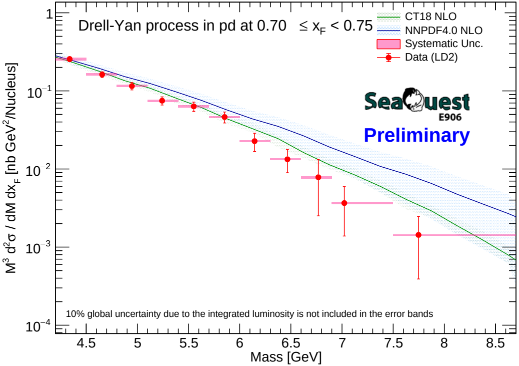


CrossSection LD2 $0.70 \leq xF < 0.75$ (Centroid)

RS67

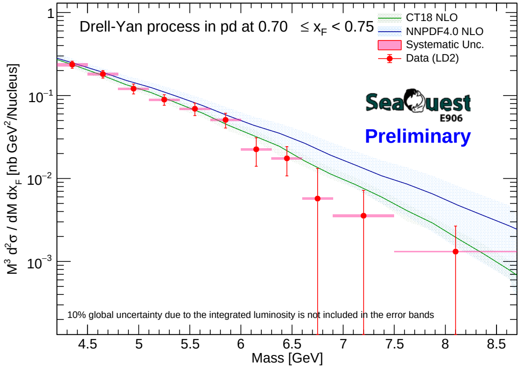


RS57-70

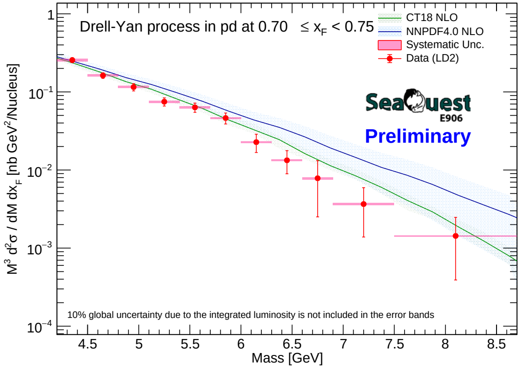


CrossSection LD2 0.70 ≤ xF < 0.75 (GeoCenter)

RS67

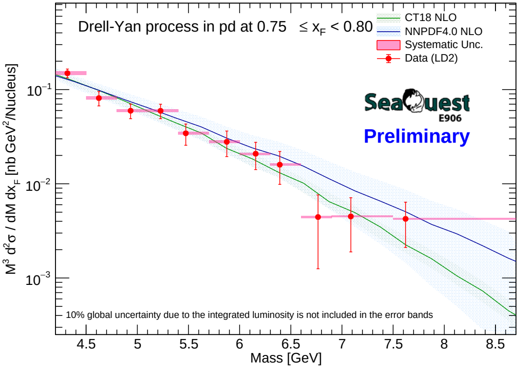


RS57-70

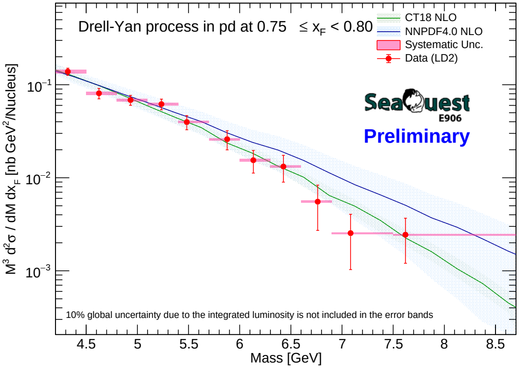


CrossSection LD2 $0.75 \leq xF < 0.80$ (Centroid)

RS67

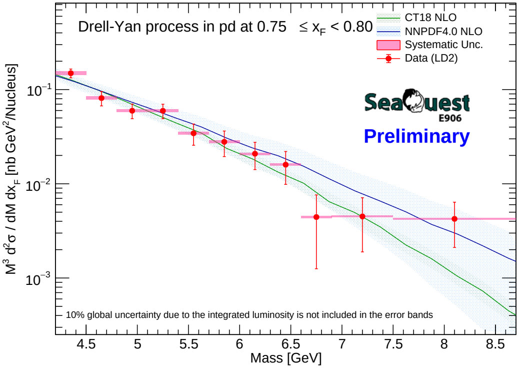


RS57-70



CrossSection LD2 $0.75 \leq xF < 0.80$ (GeoCenter)

RS67



RS57-70